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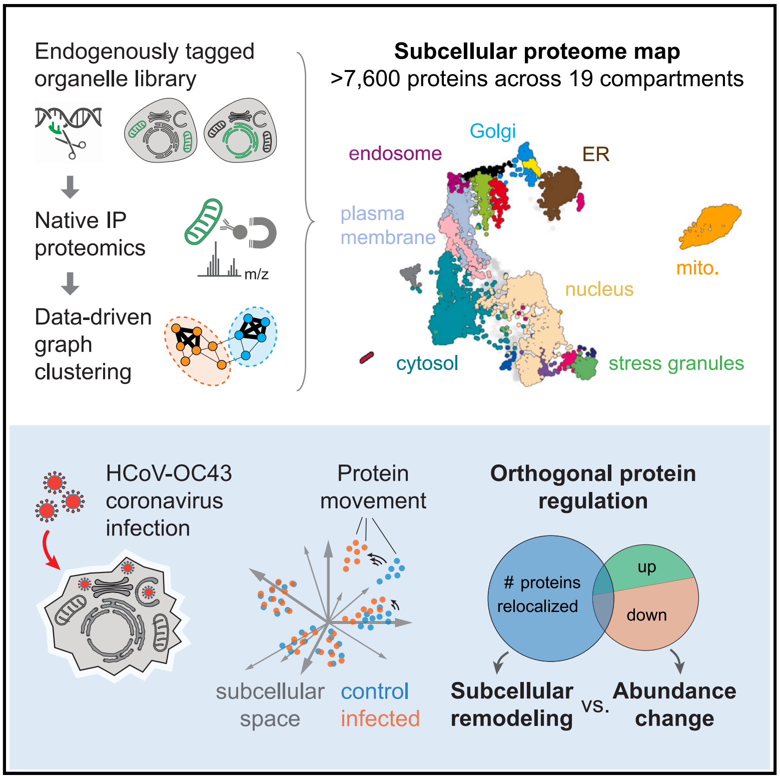
Global organelle profiling reveals subcellular localization and remodeling at proteome scale

Graphical abstract



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In brief

Organelle proteomics defines the cellular landscape of protein localization and highlights the role of subcellular remodeling in driving responses to perturbations such as viral infections.

Highlights

- Native immunoprecipitation proteomics deployed to a cell-wide organelle collection
- >7,600 proteins resolved across 19 subcellular structures and organellar interfaces
- Viral infection with HCoV-OC43 remodels global subcellular organization
- Subcellular remodeling reveals key responses not captured by abundance changes

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Global organelle profiling reveals subcellular localization and remodeling at proteome scale

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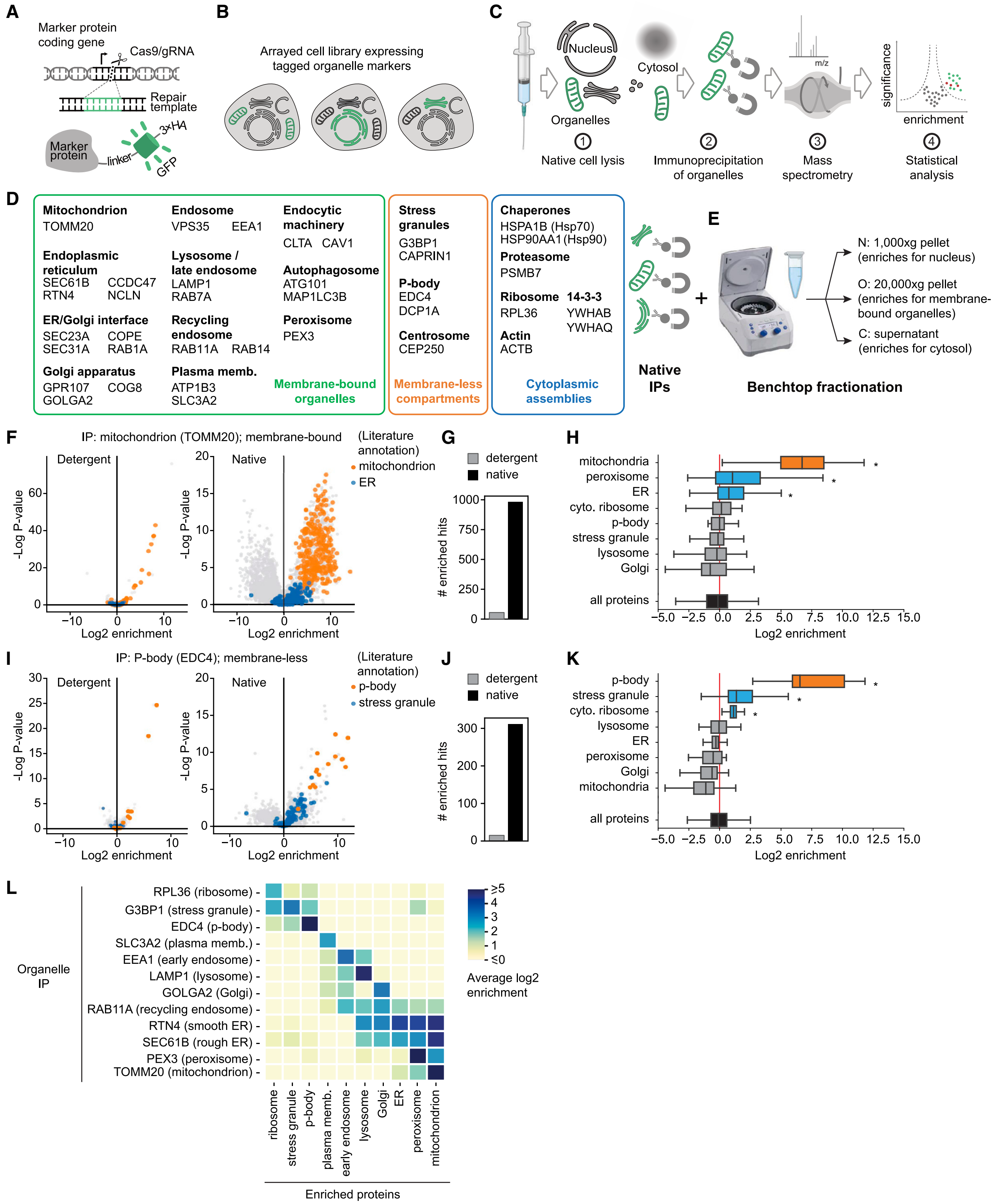
SUMMARY

Defining the subcellular distribution of all human proteins and their remodeling across cellular states remains a central goal in cell biology. Here, we present a high-resolution strategy to map subcellular organization using organelle immunocapture coupled to mass spectrometry. We apply this workflow to a cell-wide collection of membranous and membraneless compartments. A graph-based analysis assigns the subcellular localization of over 7,600 proteins, defines spatial networks, and uncovers interconnections between cellular compartments. Our approach can be deployed to comprehensively profile proteome remodeling during cellular perturbation. By characterizing the cellular landscape following HCoV-OC43 viral infection, we discover that many proteins are regulated by changes in their spatial distribution rather than by changes in abundance. Our results establish that proteome-wide analysis of subcellular remodeling provides key insights for elucidating cellular responses, uncovering an essential role for ferroptosis in OC43 infection. Our dataset can be explored at organelles.czbiohub.org.

INTRODUCTION

Internal compartmentalization is an essential feature of human cellular organization. Within the cell, membrane-bound[1,2] and membraneless organelles[3,4] provide spatial scaffolds that partition cellular functions, from oxidative ATP production in the mitochondrion to ribosome biogenesis in the nucleolus. This partitioning subdivides the cell into unique chemical environments that concentrate reactants and create the appropriate conditions for a wide range of biochemical activities.[5] Compartmentalization further enables protection from reactive intermediates and by-products (for example, peroxisomes segregate metabolic reactions that produce toxic oxidative species[6]) and provides a core mechanism to control cellular processes through the dynamic re-localization of cellular components.[5,7] The acquisition of intracellular organelles is the defining event in the evolution of eukaryotes,[8] equipping cells with drastically expanded capacities to remodel their composition and to interact with their environment.

Given the central importance of spatial partitioning for cellular functions, defining the molecular composition of organelles and how this composition is remodeled across cellular states remains a major goal in human cell biology. Accordingly, multiple methods have been developed to define the proteomes of individual cellular compartments.[9–11] Intracellular localization can be sensitively and precisely characterized by microscopy



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imaging using immunofluorescence[12] or via the expression of fluorescent protein fusions.[13,14] However, these analyses are typically carried out serially using bespoke reagents or cell lines for each protein. As a result, proteome-wide coverage remains difficult to achieve with image-based approaches, limiting their applicability to analyzing global cellular architecture and its remodeling. On the other hand, many strategies have been developed to characterize organellar proteomes and their dynamics using mass spectrometry (MS).[9,10] Local protein neighborhoods can be defined by proximity ligation using markers tagged with labeling enzymes such as APEX[15] or BioID.[16–18] The protein composition of entire compartments can be further measured following immunoprecipitation (IP),[19–26] sequential solubilization,[27,28] or biochemical fractionation. Fractionation can be used to purify a specific organelle for in-depth analysis[29] or to characterize all cellular compartments from a given sample using protein correlation profiling.[30–37] In correlation profiling, proteins from a given compartment distribute across several fractions in a manner that is distinct to that compartment.[38] Statistical methods are then used to cluster proteins that present similar enrichment patterns, thereby identifying proteins belonging to the same compartment.

Here, we present an experimental and analytical strategy for the global characterization of the subcellular proteome. We use CRISPR-Cas9-based endogenous labeling to generate a library of HEK293T human cell lines expressing epitope-tagged markers covering all major subcellular compartments, including membraneless organelles. Our approach builds upon methods originally developed for the rapid IP of lysosomes[19,22] and mitochondria[20,21] following gentle cell homogenization. We couple organelle IP with liquid chromatography-MS (LC-MS) to quantitatively define subcellular proteomes. Combining an extensive coverage of subcellular structures with a graph-based analytical framework allows us to map protein localization on a global cellular scale. In addition to assigning localization to poorly characterized proteins, our strategy defines interconnectivity networks that link adjacent compartments. Finally, we demonstrate that our method can comprehensively profile dynamic proteome remodeling upon infection with the HCoV-OC43 β-coronavirus. Our results show that many proteins are regulated by changes in their spatial distribution rather than by changes in their total abundance, establishing that proteome-wide analysis of subcellular remodeling offers key insights for the elucidation of cellular responses. In particular, we identify ferroptosis as a pro-viral response essential for OC43 infection. Overall, we provide a combination of protocols, open-source code (github.com/czbiohub-sf/Organelle_IP_analyses_and_figures), and a web-accessible data resource (organelles.czbiohub.org) for the systems-level characterization of the human proteome's architecture across cellular states.

RESULTS

IP under native conditions provides rich organellar proteomes

Whole-organelle IP enables the analysis of organellar proteomes, metabolomes, and lipidomes by LC-MS.[20–26] In a native workflow, cells are mechanically homogenized, exposing organelles while maintaining their structural integrity and internal composition.[21] Pioneering studies primarily focused on individual membrane-bound organelles, including mitochondrion,[20,21] lysosome,[22] peroxisome,[23] early endosome,[25] and Golgi complex.[26] Here, we applied this strategy to all compartments of a given cell type to obtain a comprehensive map of the subcellular human proteome (Figures 1A–1E). For each compartment, we used CRISPR-Cas9 to endogenously tag reference protein markers ("baits") in HEK293T cells, generating an arrayed library of engineered cell lines (Figures 1A and 1B). Tagging cassettes included a 102 amino-acid unstructured linker to maximize exposure of a 3xHA epitope for capture and fluorescent EGFP to allow both the selection of engineered cells by flow cytometry and the use of the tagged lines for imaging (Figure S1A). All markers were tagged on a cytosol-facing terminus, except for plasma membrane markers for which an extracellular terminus was used (Table S1). For comprehensive coverage of subcellular compartments, we tagged 37 separate markers localized to 14 membrane-bound or membraneless organelles and 5 cytoplasmic assemblies (Figure 1D). We selected markers based on three criteria: (1) markers should collectively cover all known cellular compartments,[39] including secretory[40] and recycling[41] organelles; (2) the expected localization and protein-protein interaction profiles of each marker can be maintained upon endogenous tagging, based on data from the OpenCell collection[14]; and (3) markers should be well expressed in a wide range of cell lines,[42] so that the same set might be used in different experimental contexts (Figure S1B). To increase the chance of

Figure 1. Native immunoprecipitations provide rich enrichment profiles that preserve organellar connections

(A) Endogenous tagging strategy.
(B) Arrayed cell library.
(C) Native immunoprecipitation workflow.
(D) 37 tagged markers used for a comprehensive coverage of subcellular compartments.
(E) Benchtop spin fractionation for nuclear, organellar, and cytosolic enrichment.
(F) Enrichment profiles from detergent-based (1% digitonin) vs. native IP of the mitochondrial marker TOMM20. Volcano plots show $\log_2$ enrichment of individual proteins in TOMM20 IPs relative to control (see STAR Methods). p values: t test from triplicate measurements.
(G) Enriched hits ($\log_2$ enrichment ≥ 1 and p value ≤ 0.01) for TOMM20 IPs in native vs. detergent-based conditions.
(H) Enrichment of proteins annotated to reside in different subcellular compartments in TOMM20 native IPs. In addition to mitochondrial proteins, peroxisomal and ER proteins are also significantly enriched (* $p \leq 0.05$, t test). Boxplot shows 25th/50th/75th percentiles; whiskers show 1.5× interquartile range.
(I–K) Same as (F)–(H), showing IPs for EDC4, a p-body marker (membraneless organelle).
(L) Heatmap showing $\log_2$ enrichments for proteins annotated to reside in different subcellular compartments, across different native IPs. See also Data S1. All annotation data used in Figure 1 was literature-curated (Table S2).
See also Figure S1.

efficiently capturing each organelle, we included multiple markers for some compartments (Figure 1D). We verified the subcellular localization of each tagged marker in our final library using confocal fluorescence microscopy (Figure S1C; Data S1).

Native homogenates were prepared for each marker cell line by syringing (23G needle), followed by rapid IP (10 min) on anti-hemagglutinin (HA) magnetic beads and protein quantification by label-free LC-MS (Figure 1C; STAR Methods). To capture nuclear and cytoplasmic proteins that might otherwise be excluded from the IP procedure, we used a simple centrifugation protocol to enrich nuclear (1,000 × *g* pellet), membrane-bound organellar (20,000 × *g* pellet), and cytoplasmic (supernatant) (N/O/C) fractions from an untagged homogenate (Figure 1E). All IPs and N/O/C fractionations were performed in triplicate using 10^7 cells per replicate.

Quantitative analysis revealed rich protein enrichment profiles for each organelle IP (Figures 1F–1L; Figure S1D; STAR Methods). Native IP of the mitochondrial outer-membrane protein TOMM20 showed enrichment for 982 proteins (enrichment $\geq$ 2-fold, p value $\leq$ 0.01 across triplicate experiments; Figures 1F and 1G), with the largest enrichment observed for mitochondrial proteins (Figure 1H). This was in stark contrast with the enrichment profile obtained in mild detergent- and nuclease-containing buffer[14] (Figure 1G). Native TOMM20 IP showed enrichment for proteins across all mitochondrial sub-structures (Figure S1E), verifying that our workflow preserves the structural integrity of organelles. Similar results were obtained for ER and lysosome IPs, showing comparable enrichment for membrane and lumenal components (Figure S1F). Our data further suggested that some contacts between organelles are maintained under native IP conditions. Indeed, TOMM20 IP showed modest but significant enrichment for peroxisomal and ER proteins, two organelles making direct contacts with mitochondria in the cell[43,44] (Figure 1H). By contrast, other organelles that do not contact mitochondria (e.g., lysosome or Golgi apparatus) are not seen enriched in TOMM20 IP (Figure 1H). Similar results were obtained for the IP of membraneless compartments. For example, IP of the p-body marker EDC4 showed a strong enrichment for known p-body proteins and weaker but statistically significant enrichment for proteins of stress granules or the cytosolic ribosome, two compartments known to directly interact with p-bodies[45] (Figures 1I–1K). Overall, co-enrichment was observed between compartments that have been shown to interact in the cell (Figure 1L and Data S1), suggesting a specificity in organellar co-enrichment. Enrichment plots for all IP targets are shown in Data S1 and can be explored at organelles.czbiohub.org/Native_organelle_IP_-_volcano_plots.

Graph-based analysis provides subcellular resolution to human proteome maps

We performed 58 triplicate IPs from our collection of 37 organelle markers (16 IP markers were profiled more than once). Our dataset comprises a total of 8,192 unique expressed genes (''preys,'' detected in all replicates of at least one IP; STAR Methods) and 8,538 total proteins, including isoforms of the same gene product. Each protein is defined by its enrichment across all 58 triplicate IPs and by its normalized abundance in N/O/C spin fractions (Figure 2A). This constitutes a 61-dimensional organellar enrichment profile that reflects the protein's subcellular localization. Enrichment profiles were highly correlated for proteins annotated in the literature to reside in the same organelle (Figure 2B). Reflecting the preservation of interorganellar connections in native IPs, profiles were also moderately correlated for proteins residing in compartments that directly interact in the cell (e.g., between the ER and mitochondria, off-diagonal signal in Figure 2B). The strongest profile correlation was observed between pairs of proteins engaged in high-stoichiometry interactions (Figure S2A), mirroring the intracellular co-localization of strongly interacting proteins.[14] Finally, we measured an enrichment entropy value that quantifies each protein's propensity to be enriched in multiple IPs. Proteins annotated to participate in membrane contact sites[46] displayed higher entropy than non-contact site proteins (Figure S2B), indicative of their enrichment in multiple organelles.

To further exploit the rich signatures contained in each protein's enrichment profile, we developed an analytical strategy inspired by single-cell RNA sequencing (scRNA-seq). There, relationships between cells are typically represented in a *k*-nearest neighbor (k-NN) graph in which edges correspond to the similarity of two individual cells in gene expression space, considering edges with a number *k* of most-similar neighbors for each cell. This graph is the foundation for dimensionality reduction algorithms such as UMAP[47] and, separately, for unsupervised clustering using the Leiden algorithm frequently used for the annotation of cell types.[48] This reflects the power of graph-based representation to encapsulate complex relationships from high-dimensional data.[49]

We constructed a k-NN graph connecting each of the 8,538 proteins in our dataset to its 20 nearest neighbors, with edge weights derived from the Euclidean distances between 2 proteins in the 61-dimensional enrichment space (Figure 2C; STAR Methods). To identify proteins from the same subcellular compartment, we performed unsupervised graph-based clustering using the Leiden algorithm.[48] We then objectively labeled each protein cluster according to its most enriched Gene Ontology term using the COMPARTMENTS[39] and GO-*Cellular component*[50] databases and merged clusters sharing the same ontology label. We finally refined annotations using local neighborhood consensus: a protein is annotated with the cluster-based label most found amongst its direct neighbors (Figure 2C). This step acts to better define boundaries between clusters. Complete enrichment and annotation data can be found in Table S3. We obtained definitive ontology-based annotation for 7,666 proteins (90% of the dataset), leaving 875 unclassified. The latter mostly correspond to soluble proteins that lack a clear enrichment in N/O/C spin fractions (Figure S2C), reflecting a broad multi-localization between nuclear, cytosolic, and membrane compartments.

Figure 2D shows a two-dimensional UMAP derived from this analysis, representing a spatial map of the HEK293T proteome (see also Table S3 and organelles.czbiohub.org/Subcellular_UMAP). We distinguished 20 separate subcellular compartments, with sizes ranging from 18 (p-body) to 1,836 proteins (cytosol) (Data S1). The arrangement of compartments within the map follows known biological relationships; for example,

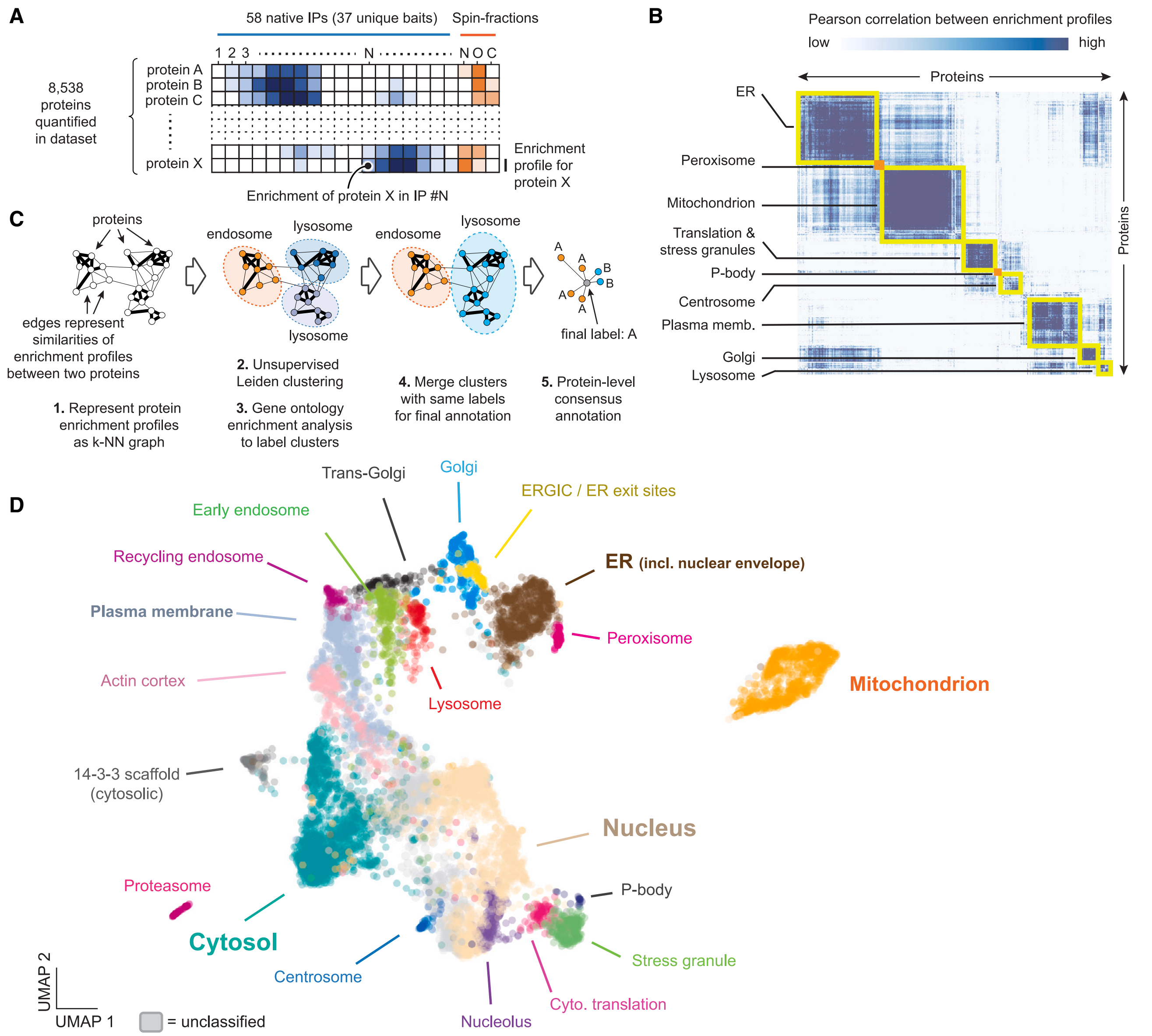

Figure 2. Graph-based analysis provides a subcellular map of the human proteome

(A) Overall data structure. Our dataset identifies 8,538 proteins defined by their enrichment across individual native organelle IPs and spin fractions.

(B) Pairwise Pearson correlation between enrichment profiles of literature-curated marker proteins for each compartment (Table S2). High correlation is observed between proteins from the same compartment (diagonal highlights) and, to a lesser extent, between proteins in compartments known to interact in the cell (for example, ER and mitochondrion).

(C) Graph-based strategy for data-driven annotation of protein localization. The enrichment matrix is transformed into a k-NN graph (1), from which co-enriched proteins are identified using Leiden clustering (2). Gene Ontology enrichment allows the annotation of clusters corresponding to separate subcellular compartments (3 and 4). A final regularization step assigns to each protein the label most represented amongst its direct neighbors (5).

(D) Two-dimensional UMAP showing the 20 subcellular compartments identified by the enrichment analysis in (C). Each dot represents an individual protein.
See also Figures S2 and S3.

membrane-bound organelles are arranged in a sequence that matches the cell's secretory pathway from ER to Golgi to endosomes to plasma membrane (Figure 2D). In addition to classically defined compartments, our annotations also include a well-defined cluster of 130 proteins ("14-3-3 scaffold") defined by their strong enrichment in IPs of the 14-3-3 proteins YWHAB and YWHAQ. 73% of proteins in that cluster were also identified in a recent proteomics study of 14-3-3 clients in HEK293T cells, supporting the annotation.[51] Re-calculating the subcellular map without the YWHAB and YWHAQ IPs demonstrated that the majority of 14-3-3 scaffold proteins are cytosolic (Figure S2E; Table S3).

Validation and comparison to existing datasets

To validate annotations using an orthogonal approach, we trained an XGBoost[52] supervised machine learning classifier to predict the organelle membership of individual proteins based on reference data (Table S4). We limited classification to 15 compartments for which enough literature-curated reference data was available (Table S2; excluding, for example, *trans*-Golgi or recycling endosome, which most other datasets do not distinguish). We then performed receiver operating characteristic (ROC) analyses to quantify the agreement between graph- and classifier-based annotations (Figure S2D). High values of area under the ROC curves ($\geq$0.93) were obtained in all cases, supporting our graph-based annotations. Graph-based annotations were further supported by a comparison with image-based annotations curated in the OpenCell project[14] (Figure S2F; Data S1; Table S5).

Next, we compared our annotations with those of proteome-scale subcellular MS datasets that use a variety of fractionation, ultracentrifugation, and proximity ligation methods.[18,27,36,37,53] Our graph-based annotations disambiguated a higher number of subcellular compartments than all other studies (Figure S3A, especially among secretory, recycling, and membraneless compartments), while also covering a larger fraction of the proteome (Figures S3B and S3C). All annotations across datasets are summarized in Table S5 (see also Data S1). Because there exists no objective ground truth for protein subcellular localization, discrepancies between datasets are hard to resolve. The mitochondrion stands out, however, as a compartment for which extensive curation is available. In particular, the MitoCarta[54] and MitoCoP[29] datasets curate multi-modal data sources across cell types and cell states to define the mitochondrial proteome. Combining MitoCarta and MitoCoP to define an objective ground truth, we performed a precision/recall analysis on the lists of mitochondrial proteins annotated in different proteome-scale datasets (Figure S3D; Table S5). Our annotations achieved the highest recall across all datasets (77%, 1.45× higher than the average recall) while being one of only two datasets to exhibit an estimated false-discovery rate lower than 5% (3.2%).

To further compare the subcellular resolving power of different MS-based approaches within a consistent framework, we reanalyzed published data using the k-NN graph strategy described above. For each dataset, we computed a two-dimensional UMAP (Data S1) and calculated scores that measure the extent to which proteins sharing a given ground-truth label form well-delineated clusters in these maps (STAR Methods). We measured two clustering scores capturing protein co-localization at different scales: an organelle-level score, using annotations curated from the literature (Table S2), and a protein-complex score, using annotations from CORUM.[55] Our dataset achieved higher clustering scores than other methods (Figure S3E). Computing clustering scores also allowed us to quantify how the subcellular resolution power of our IP-based approach varies as different numbers of IPs are included. When using a reduced set of 19 IPs limited to one IP per subcellular compartment, we observed a modest reduction of clustering scores (Figures S3E and S3F) while still achieving higher scores than other methods.

Subcellular protein networks define functional signatures and quantify cellular organelle connections

The k-NN graph representation of our dataset (hereafter, the "k-NN map") contains information that extends beyond organellar membership. Figures 3A and 3B show the k-NN map centered on WASHC5 (Strumpellin), a subunit of the actin-nucleating WASH complex (WASHC). WASHC regulates cargo transport from endosomes to *trans*-Golgi network (TGN), plasma membrane, or lysosomes.[56–58] Reflecting its function, WASHC5 localizes at the nexus between endosomal, TGN, and lysosomal clusters in the k-NN map (Figure 3A). Furthermore, WASHC performs its role by associating separately with the Retromer and Commander complexes,[58–60] which both control endosomal tubulation but regulate trafficking of different cargo. Recapitulating these complex relationships, WASHC subunits are connected to both Retromer and Commander in the k-NN map, while Retromer and Commander occupy distinct territories (Figure 3B). In another example, the nuclear pore subunit NUP205 is positioned at the interface between nuclear and ER clusters in the k-NN map (Data S1), together with other nuclear pore subunits and key components of the nuclear lamina or nuclear import machinery. Nuclear envelope and ER are contiguous in the cell and are not directly resolved in our native IP workflow: upon gentle homogenization, we expect the two compartments to remain physically connected. However, the k-NN map contains information to resolve these adjoining compartments. Altogether, these examples illustrate how the detailed structure of the k-NN map can be mined for functional exploration. Protein k-NN connections are compiled in Table S6 and can be explored at organelles.czbiohub.org/Protein_network_graph.

Interorganellar communication is an essential part of cellular physiology.[2] We quantified connectivity densities defined by the number of k-NN connections bridging two annotation clusters normalized by cluster size. This revealed known functional relationships between organelles, with the strongest connections bridging ER-Golgi intermediate compartment (ERGIC) and Golgi, plasma membrane and actin cortex, or the translation machinery and stress granules (Figure 3C). Many proteins act at the interface between organelles, ensuring cargo transport, signal propagation, or metabolic coordination. To further annotate organellar connectivity, we identified "interfacial" proteins in our k-NN map. We adopted a version of the Jaccard coefficient used in graph theory to quantify a given node's propensity to make connections with two separate clusters (STAR Methods). We quantified the distribution of the Jaccard coefficient for all proteins using graph-based annotation clusters (Figures 3D and 3E). Values were low for most proteins (median = 0.0075), with a long tail of proteins of higher coefficients (Figure 3D). We defined interfacial proteins as the high-value outliers in this distribution (values above third quartile + 1.5× interquartile range; Figure 3D). The list of 950 interfacial proteins (Table S6) includes an over-representation of interfaces between organelles sharing high connectivity densities (e.g., Golgi/ERGIC or plasma membrane/actin cortex, compare Figures 3C and 3E).

Our data-driven analysis correctly identified many proteins specializing in interfacial transport or regulation within the vesicular system (Figure 3F). These include AP1, AP2, and AP3 coat proteins, which scaffold clathrin-mediated transport at the

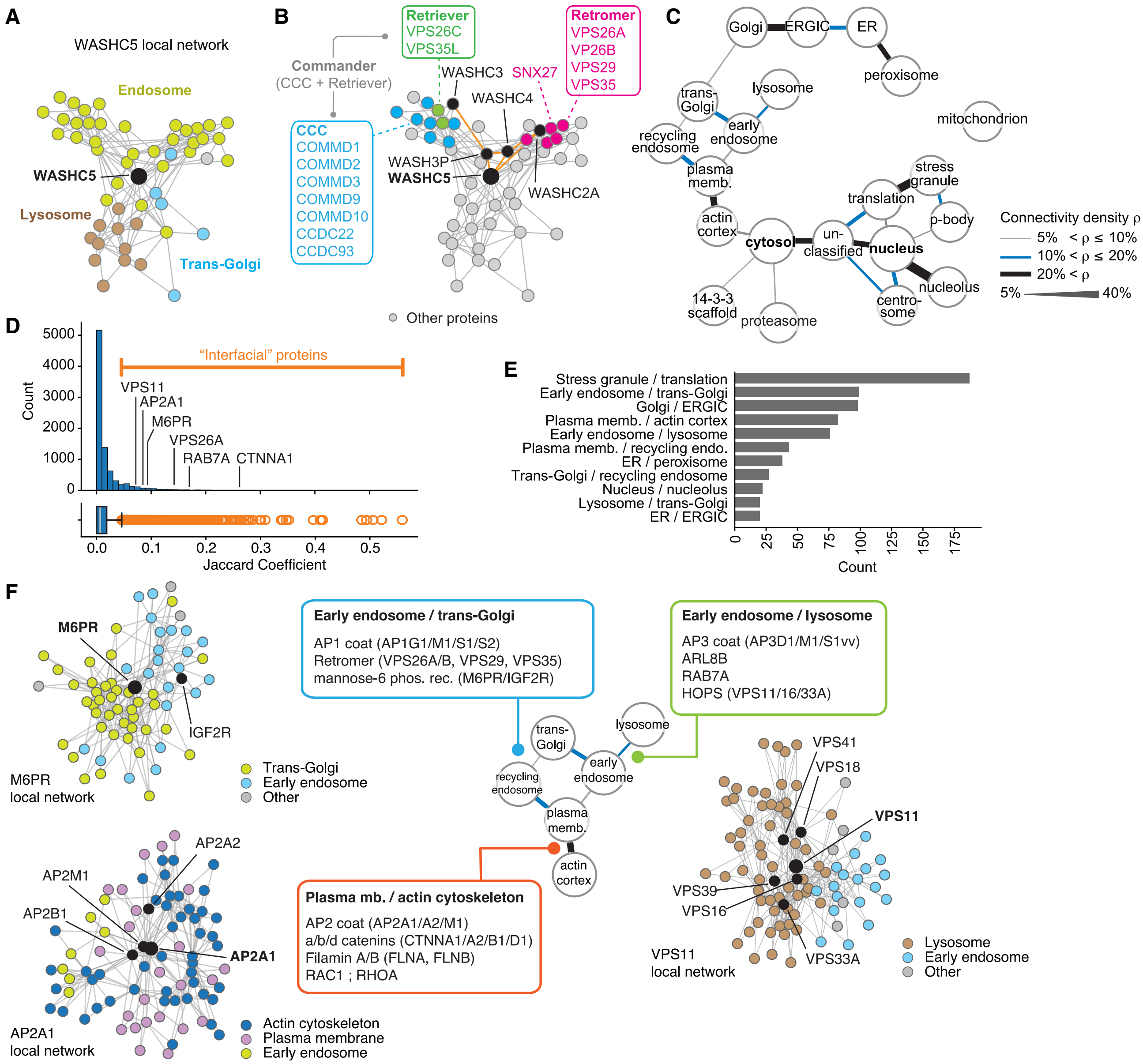

Figure 3. Protein spatial networks identify co-functioning proteins and organelle interfaces

(A) Local k-NN network centered on WASHC5, a subunit of the WASH complex that regulates protein trafficking between endosomes, lysosomes, and *trans*-Golgi network.

(B) Annotated WASHC5 k-NN network, highlighting individual subunits for the WASH, Retromer, and Commander protein complexes.

(C) Density of k-NN connectivity between organelle clusters quantifies intercompartment crosstalk. Clusters are arranged to replicate the arrangement in Figure 2D.

(D) Interfacial proteins are defined as high-value outliers in the distribution of the Jaccard coefficient in the entire dataset. Boxplot shows 25th/50th/75th percentiles; whiskers show 1.5× interquartile range.

(E) Interfaces most represented in the set of interfacial proteins.

(F) Examples of interfacial proteins in the vesicular pathway. See text for details.

endosome/TGN, plasma membrane/actin cortex, and endosome/lysosome interfaces, respectively[61,62] (Figure 3F, all panels). The local k-NN network around AP2 subunits mirrors the central role of the actin cortex in trafficking endocytic plasma membrane cargo toward the endosomal system[62] (Figure 3F, bottom left). Other examples include (1) the M6PR and IGF2R mannose-6-phosphate receptors,[63] and the Retromer complex,[58] which mediate recycling between endosome and *trans*-Golgi (Figure 3F, top left); (2) at the endosome/lysosome interface, the HOPS complex[64] which tethers endosomes and lysosomes, and the small GTPases ARL8B[65] and RAB7A[66] (Figure 3F, right); (3) the catenin[67] (CTNN) and filamin[68] (FLN) protein families that play a key role in bridging the actin cytoskeleton to plasma membrane effectors, an interface that also includes the small GTPases RAC1 and RHOA which regulate membrane actin polymerization to control cell migration[69] (Figure 3F, bottom left).

Annotating protein subcellular localization

Annotations of subcellular localization are summarized in Table S4. We report results from graph-based clustering analysis (including interfacial proteins), together with high-confidence (classification probability > 80%) predictions from the XGBoost machine learning classifier. The two strategies provide complementary insights for the comprehensive curation of protein localization. Unsupervised graph-based clustering enables *de novo* annotations, even when no critical mass of reference data is available to train a classifier (for example, allowing us to resolve TGN or recycling endosomes). On the other hand, classifier-based methods might be more precise in cases where sufficient ground truth is available. Using the mitochondrial proteome as a reference, graph-based annotations supported a higher recall than classifier-based annotations (77% vs. 69%), at the expense of a slightly increased false-discovery rate (3.2% vs. 1.5%). Another defining advantage of graph-based methods is to provide an explicit framework to identify proteins acting at the interface between compartments. This produces a layer of annotation not captured by classification that only assigns a single label for each protein.

Our final annotation set includes graph-based annotations for 7,666 proteins (90% of the dataset) and high-confidence classifier-based annotations for 6,348 proteins (74% of the dataset). Across the 15 compartments defined in both sets, graph- and classifier-based annotations shared an 89% match (Data S1).

De-orphaning protein localization

To validate annotations, we performed a microscopy-based analysis to de-orphan the localization of poorly characterized membrane-related proteins (Figure 4A). We identified 82 "orphan" transmembrane or lipidated proteins in our dataset by filtering for proteins with minimal annotation in the COMPARTMENTS database[39] (score $\leq$ 3, indicating low-confidence or absent annotations) that also had no "subcellular location" specified in Uniprot.[70] We used CRISPR-Cas9 to endogenously tag all 82 proteins with a split-mNeonGreen fluorescent cassette separately at their N or C terminus (Table S1). 31 cell lines exhibited enough fluorescent signal to be selected by flow cytometry (correlated with higher protein abundance, Figure S4A) and were imaged using live-cell confocal fluorescence microscopy (Figure 4B).

To quantitatively characterize subcellular distributions, we used the *cytoself* deep-learning model to encode protein localization images into latent space embeddings[71] and compared the 31 orphan proteins to organelle markers from the OpenCell collection[14] (Figure S4B). The Pearson correlation between latent space embeddings of each orphan vs. marker protein pair quantifies the similarity between their subcellular distribution. Similarity scores above 0.6 reflect a close match between subcellular distributions,[14] and a high score between an orphan protein and a marker for a given organelle validates its localization. For example, the highest similarity score for the orphan protein NAT14 was with the ER protein CERS5 (score: 0.76), confirming NAT14's annotation as an ER protein (Figure S4C). High-scoring matches were found for 22 out of the 31 orphan proteins, validating their annotated localization (Figure S4D). Of these, TMEM209 localizes to the nuclear membrane, which is not explicitly distinguished from the ER by native IP. TMEM209's closest neighbors in the k-NN map are known components of the nuclear membrane (Figure S4E), highlighting how k-NN information can help refine functional annotation. Separately, we confirmed the localization of the orphan proteins SCAMP3, CMTM4, CMTM7, SCAMP4, SLC35F1, and SLC35F2 to recycling endosomes by measuring their co-localization with RAB14, a marker of recycling compartments[72,73] (Figure S4F). No high-scoring match could be identified for SFT2D3, KIAA1522, and MPZL1, but their annotated localizations were supported by literature data. SFT2D3 (Golgi) and MPZL1 (plasma membrane) were localized in the corresponding compartments in subcellular proteomics studies using ultracentrifugation[36] and proximity ligation,[18] and KIAA1522 (actin cortex) is a direct interactor of multiple WAVE complex subunits[14] regulating actin dynamics at the plasma membrane.[74]

Overall, the imaged subcellular distribution of the tagged orphan proteins matched our proteomics-based annotations in 29 out of 31 cases (94% match, Figures 4B and 4C). Among the remaining two cases, TPRA1 displayed a non-uniform distribution at the lysosomal surface while being annotated as interfacial TGN/recycling endosome in our dataset. Finally, endogenously tagged GLT8D1 displayed a nucleolar localization despite being annotated as a Golgi/TGN protein. A nucleolar localization would be unexpected for a type-II transmembrane protein such as GLT8D1. Moreover, GLT8D1 localization to the Golgi is supported by literature data.[75,76] Therefore, we hypothesize that the nucleolar localization we observe might be a tagging artifact.

Pan-cellular remodeling during HCoV-OC43 infection

To capture the global intracellular remodeling during a perturbation, we profiled cells infected with HCoV-OC43, an enveloped, positive-strand RNA β-coronavirus. Typically causing mild respiratory symptoms in humans,[77] OC43 belongs to the same family as SARS-CoV-1 and SARS-CoV-2 but only requires biosafety level-2 containment. We infected HEK293T at a multiplicity of infection (MOI) of 0.25 and profiled their proteome 48 h post infection (48 hpi; >80% infected cells: Figure S5A). We selected 25 organelle-tagged cell lines (Figure S5B) for IP and N/O/C spin fractionation in uninfected vs. infected conditions, followed by LC-MS. For both conditions, the protein enrichment matrix was translated into a k-NN graph (Figure 5A). We then used the aligned-UMAP algorithm[47] to project graph data into a shared 10-dimensional Euclidean space from which distances between their infected vs. uninfected conditions could be calculated (Figure 5A; STAR Methods). For each protein, this distance represents a subcellular remodeling score that measures that protein's repositioning in the underlying k-NN maps upon OC43 infection. The overall distribution of remodeling scores showed that most proteins had low scores (indicating an overall conservation of their k-NN neighborhoods), with a long tail of high-value outliers (Figure 5B; outliers defined above third

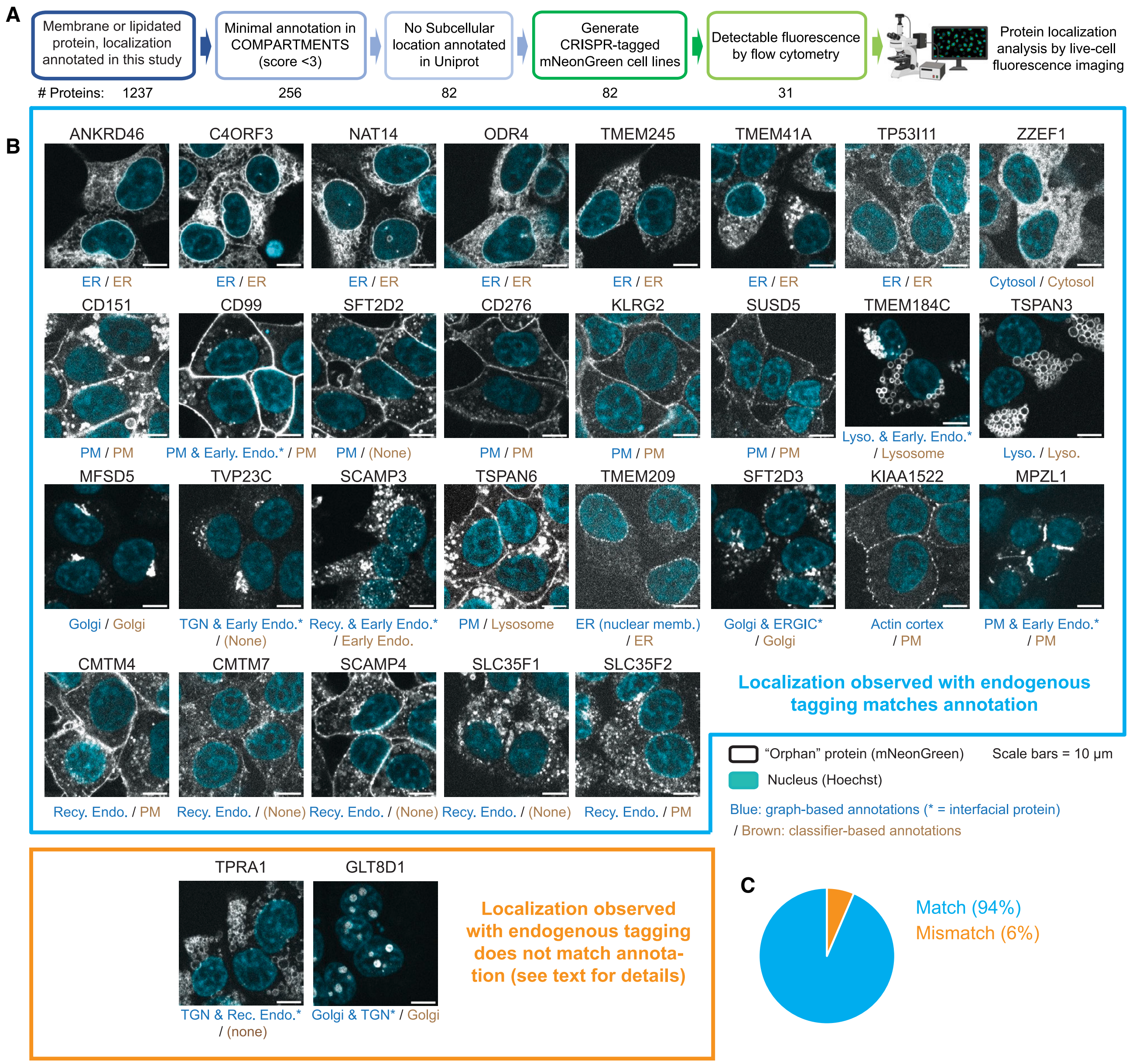

Figure 4. De-orphaning subcellular localization

(A) Strategy to identify and characterize proteins with minimal annotations for subcellular localization in public databases.

(B) Representative images for 31 final protein targets, endogenously tagged with split-mNeonGreen in HEK293T and characterized by confocal fluorescence microscopy. Graph-based (blue) and classifier-based (brown) localization annotations are reported. Asterisks denote interfacial proteins. Abbreviations are as follows: ER, endoplasmic reticulum; ERGIC, ER-Golgi intermediate compartment; PM, plasma membrane; TGN, *trans*-Golgi network.

(C) Aggregate results: endogenous tagging supports the annotations of subcellular localization in 94% of cases (29 out of 31 targets).

See also Figure S4.

quartile + 1.5× interquartile range). These 633 outliers define "infection hits" (Table S7): proteins whose subcellular environment changes significantly during infection. The infection hits encompass 8% of the proteome measured in our dataset, indicating the large extent of pan-cellular remodeling occurring upon viral infection. To more easily visualize how individual infection hits distribute across subcellular compartments, we produced aligned two-dimensional UMAPs under uninfected and infected conditions. A movie of morphed trajectories between conditions captures the remodeling behavior of individual proteins (Video S1). We performed low-resolution Leiden clustering to separate the UMAPs into 10 broad subcellular territories (Figures 5C–5E; Table S7). Infection hits were found in all territories and exhibited a wide variety of remodeling behaviors (Figure 5E, diagonal trajectories). The infected map also revealed the subcellular localization of 21 virally expressed proteins

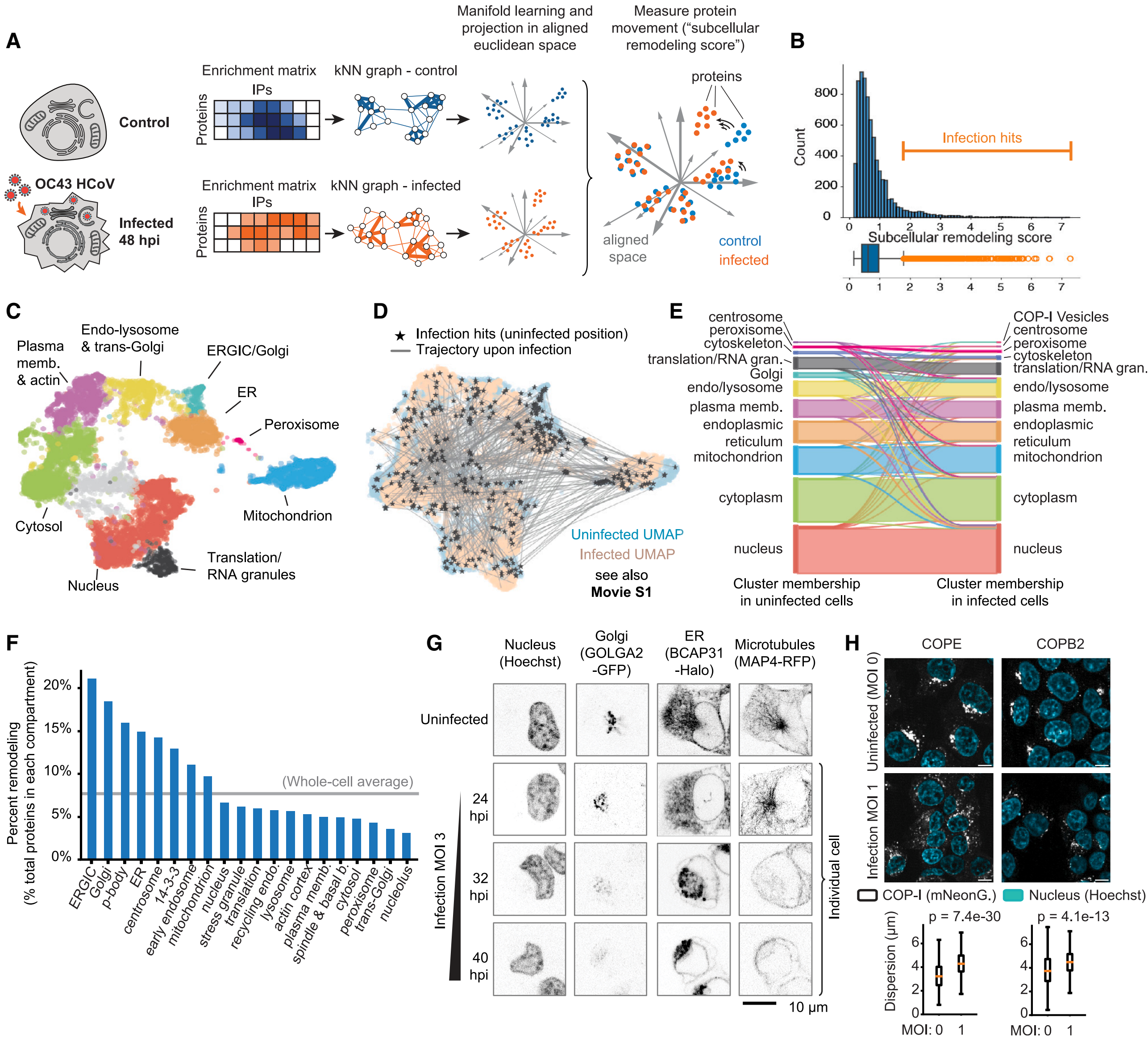

Figure 5. Pan-cellular remodeling during HCoV-OC43 infection

(A) Strategy for graph-based quantification of subcellular remodeling. A score is quantified for each protein, representing remodeling of local subcellular environment. See text for details.

(B) Distribution of subcellular remodeling scores across the dataset. Interfacial proteins are defined as high-value outliers in the distribution. Boxplot shows $25^{th}/50^{th}/75^{th}$ percentiles; whiskers show 1.5× interquartile range.

(C) 2D UMAP of uninfected organelle dataset showing subcellular territories defined by low-resolution Leiden clustering.

(D) Aligned uninfected (blue) and infected (salmon) UMAPs showing individual 2D trajectories for infection hits. Black star: original position in uninfected UMAP. See also Video S1.

(E) Sankey diagram showing the distribution of infection hits (diagonal lines) across subcellular territories between uninfected and infected conditions. Related to (C) and (D).

(F) Proteome remodeling within individual compartments, showing the fraction of infection hits over total number of proteins in each compartment.

(G) Live-cell confocal microscopy imaging from endogenously tagged cell lines shows significant remodeling of Golgi, ER, and microtubule network upon OC43 infection.

(H) Live-cell confocal microscopy imaging of endogenously tagged COP-I subunits. Fluorescent signals concentrate in a perinuclear region in uninfected cells but are dispersed across the cell upon infection (MOI = 1; 48 hpi). Distance between COP-I fluorescence signal and the nuclear boundary in individual cells is used to quantify dispersion (p value: Student's t test). Boxplots show 25^{th}, 50^{th}, and 75^{th} percentiles; whiskers show 1.5× interquartile range.

See also Figure S5.

(Figures S5C and S5D). Most viral proteins were annotated as ER-localized (Figure S5D), including replication-associated proteins[78,79] such as the RNA polymerase RDRP, the viral helicase HEL, and non-structural proteins NS8, NS9, and NS10. This mirrors the localization of the OC43 replication compartment in ER-adjacent double-membrane vesicles.[78,79]

Quantifying the fraction of infection hits within individual organelles provides a finer picture of subcellular reorganization (Figure 5F). ERGIC, Golgi, centrosome, and ER exhibited some of the highest degrees of remodeling (Figure 5F). This matches the known cell biology of coronavirus replication and assembly at the ER/Golgi interface: structural proteins are first inserted into the ER membrane and transit to the ERGIC to reach complex replication/assembly compartments where mature virions are formed.[79,80] Viral particles then traffic to the Golgi apparatus for glycosylation and other post-translational modifications (PTMs).[80] We verified the pronounced remodeling of the Golgi, ER, and centrosome using live-cell microscopy. HEK293T cells were endogenously tagged with fluorescent reporters on three separate markers: GOLGA2 (GM130, *cis*-Golgi), BCAP31 (ER), and MAP4 (microtubule cytoskeleton). Upon infection, we observed a striking fragmentation of the Golgi accompanied by a condensation of the ER (Figure 5G). This also coincided with large changes in the distribution of microtubules in infected cells, compatible with a loss of the microtubule-organizing center following centrosomal remodeling. In SARS-CoV-2, the NSP13 viral protein directly interacts with host proteins involved in both centrosomal and Golgi organization, suggesting a possible link between the remodeling of both compartments.[81] While an in-depth elucidation of the mechanisms driving ER/Golgi remodeling upon OC43 infection is beyond our current scope, we observed that the distance between Golgi and lysosomal proteins in organellar enrichment space decreased significantly during infection (Figure S5E). This might mirror the unconventional lysosomal exocytosis route that has been proposed for β-coronaviruses egress, believed to involve direct protein transport from Golgi to lysosomes.[80] In addition, all subunits of the COP-I complex (a vesicular coatamer that mediates cargo transport inside the Golgi apparatus and between Golgi and ER) were identified as infection hits (Figure S5F). Fluorescence imaging of endogenously tagged COPE and COPB2 subunits revealed their subcellular redistribution upon OC43 infection: subunits were concentrated in a perinuclear region in uninfected cells but found dispersed across the cell upon infection (Figure 5H). COP-I regulates the trafficking of the SARS-CoV-2 spike glycoprotein,[82,83] and COPB2 knockdown was recently shown to significantly decrease SARS-CoV-2 viral titers,[84] suggesting an important role for COP-I in the β-coronavirus life cycle.

Subcellular remodeling reveals cellular responses not captured by abundance changes and identifies ferroptosis as an essential pro-viral response

To contextualize our organelle profiling results with respect to global gene expression, we measured whole-cell abundance changes of transcripts and proteins in OC43-infected vs. control conditions (Figures S6A and S6B; Table S7). We observed remarkably little overlap between infection hits defined from subcellular remodeling and differentially abundant proteins or transcripts (Figures 6A and 6B; Figures S6A and S6B). This suggests orthogonal modes of functional regulation: many proteins are dynamically regulated at the level of their spatial environment rather than by changes in cellular abundance. Vice versa, many proteins are regulated at the abundance level, while their subcellular niche stays the same. Applying pathway enrichment[85,86] to the different groups of infection-regulated proteins or transcripts demonstrated that profiling subcellular remodeling enables the identification of cellular responses not captured by abundance changes (Figures 6A and 6B). Specifically, subcellular infection hits were enriched for proteins involved in the metabolism of glycosaminoglycans (GAGs, a group of surface-exposed polysaccharides that includes heparan sulfate glycans), as well as in ferroptosis, autophagy, and cellular senescence (Figure 6A). Multiple lines of evidence support the direct relevance of these responses to the cell biology of OC43 infection: (1) GAGs are the surface receptor for OC43 entry under cell culture conditions,[87] so that modulation of their metabolism (which remains poorly understood[88]) might be beneficial for the viral life cycle, or for innate immune responses; (2) autophagy plays an important role in the replication and pathogenesis of multiple coronaviruses,[89,90] including OC43[91]; (3) senescence is a primary host-cell response to infection by many viruses, for example, SARS-CoV-2.[92]

Ferroptosis was among the most highly enriched processes among subcellular remodeling infection hits (Figure 6A). Ferroptosis is a form of non-apoptotic and non-necrotic programmed cell death dependent on cellular iron,[93,94] that was recently implicated in the regulation of replication for OC43 and other positive-strand RNA viruses.[95] During ferroptosis, iron (primarily Fe^{2+}) catalyzes unrestrained lipid peroxidation, leading to oxidative stress and to a gradual destruction of intracellular membranes.[94] Under normal conditions, intracellular iron is stored in ferritin, a high-molecular weight protein cage. The autophagy-mediated degradation of ferritin (ferritinophagy) leads to the release of free iron in the cytoplasm and drives ferroptosis.[96–98] Multiple protein pathways play a central role in the induction and regulation of ferroptosis in cells, including ACSL4 (an acyl-transferase that incorporates polyunsaturated fatty acids into lipids, generating the substrates for peroxidation[99]), GPX4 (a peroxidase that clears toxic peroxides, counteracting ferroptosis[94]), and NCOA4 (the autophagy adapter for ferritinophagy[100]).

We validated the induction of ferroptosis in our model: OC43 infection led to an increase of intracellular Fe^{2+} as detected by the fluorescent probe FeRhoNox-1, which could be reversed by the iron chelator nitroxoline (Figure 6C; Figure S6C). Strikingly, inhibition of ferroptosis with liproxstatin-1[101] or ferrostatin-1[101] at non-toxic concentrations led to a significant decrease of infection, while activation of ferroptosis with RSL3[102] increased infection rates (Figure 6D; Figure S6D). These results establish that ferroptosis is both pro-viral and essential for OC43 infection in human HEK293T cells. Ferrostatin-1 was recently shown to also inhibit OC43 replication in rhesus macaque LLC-MK2 cells.[95] While follow-up work will be needed to better characterize the mechanistic interplay between ferroptosis and the viral life cycle, our data provides insights into some aspects of this mechanism. ACSL4, GPX4, and NCOA4 all exhibited high subcellular remodeling scores (Figure 6E), suggesting

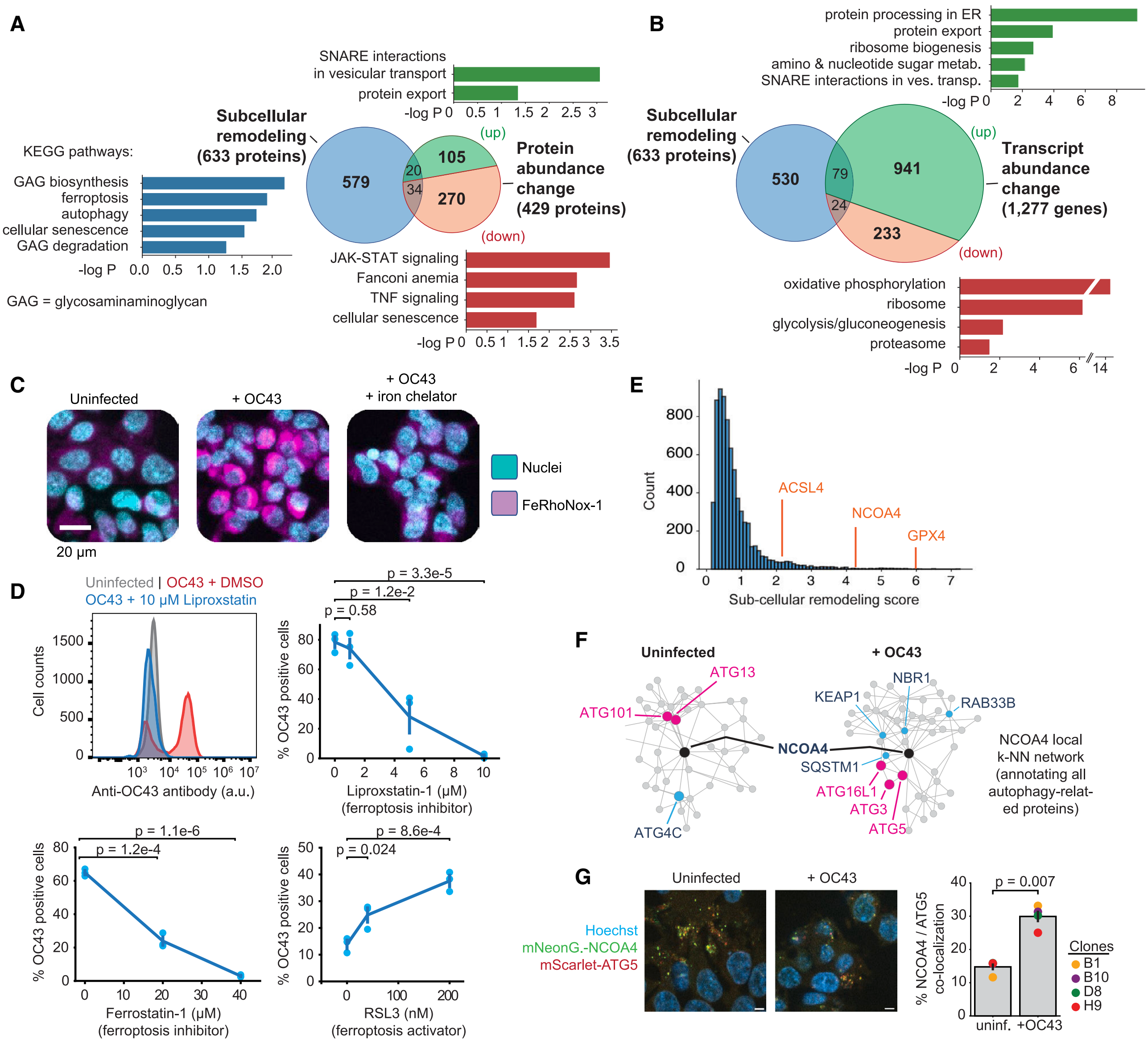

Figure 6. Subcellular remodeling upon OC43 infection reveals cellular responses not captured by abundance changes

(A) KEGG pathway enrichment analysis of infection hits from subcellular remodeling (blue) vs. proteins that exhibit significant changes in whole-cell abundance (including 125 up-regulated proteins, green, and 304 down-regulated proteins, red). *p* values: Student's t test.

(B) Same as (A), showing differentially regulated transcripts (including 1,020 up-regulated transcripts, green, and 257 down-regulated transcripts, red). *p* values: Student's t test.

(C) Ferroptosis induction in OC43-infected cells (MOI = 1; 48 hpi). Cells were treated with 5 μM FeRhoNox-1 (magenta) to detect ferroptosis. Nitroxoline (12 μM) was used to chelate Fe^{2+} ions.

(D) Detection of OC43-infected cells by flow cytometry (anti-OC43 antibody 541-8F). Treatment with ferroptosis inhibitors liproxstatin-1 and ferrostatin-1 induced a dose-dependent decrease in infection (MOI = 1; 48 hpi). Conversely, treatment with the ferroptosis activator RSL3 induced a dose-dependent increase in infection. For RSL3 experiments, a low MOI was used to enable a large infection dynamic range (MOI = 0.05; 48 hpi). *p* values: Student's t test. Symbols show independent triplicate experiments, bars show SEM.

(E) Subcellular remodeling scores of ACSL4, NCOA4, and GPX4 (cf. Figure 5B).

(F) Local k-NN networks centered on NCOA4 in both infected and uninfected conditions. All autophagy-related proteins are highlighted. See Figure S6F for fully annotated networks.

(G) Co-localization between NCOA4 and ATG5 significantly increases upon OC43 infection (MOI = 1; 48 hpi). Four separate cell clones harboring dual CRISPR-mediated endogenous tags (mNeonGreen-NCOA4; mScarlet-ATG5) were compared. In each clone, the percentage of mNeonGreen-positive pixels that are also mScarlet-positive measures co-localization. *p* value: Student's t test.

See also Figure S6.

that multiple ferroptosis pathways might be regulated upon infection. This remodeling was not accompanied by changes in protein abundance (Figure S6E). Furthermore, local k-NN networks revealed a specific shift in NCOA4's subcellular environment upon infection (Figures 6F and S6F). In uninfected cells, NCOA4's nearest neighbors include ATG101 and ATG13, part of the ULK kinase complex that regulates pre-autophagosomal structures and autophagic initiation.[103] By contrast, NCOA4's nearest neighbors upon infection include ATG5 and ATG16L1, key components of the ATG12-conjugation system that mark active, elongating autophagosomes.[103] This suggested an increased association between NCOA4 and active autophagosomes upon infection, which we verified by imaging the colocalization between NCOA4 and ATG5 in endogenously tagged cell lines (Figure 6G). This establishes a role for active ferritinophagy in the cell biology of OC43 infection.

Altogether, our analysis demonstrates how profiling subcellular remodeling can complement whole-cell proteomics or transcriptomics assays to capture the full landscape of cellular responses triggered by a given perturbation.

Interactive data sharing at organelles.czbiohub.org

To facilitate access and exploration, we built an interactive web application at organelles.czbiohub.org. This portal provides an intuitive interface to explore and download enrichment data from individual organelle IPs, protein-level k-NN networks, subcellular localization UMAPs, and subcellular remodeling induced by OC43 infection (Figure 7).

DISCUSSION

Here, we described an integrated experimental and analytical framework to advance the characterization of the subcellular proteome and its dynamics. We combined three main features. First, we applied native IP workflows to a cell-wide collection of organelle markers. Second, we developed a graph-based formalism to define functional relationships between proteins while annotating their subcellular localization. Third, we extended this strategy to profile proteome dynamics and identify proteins whose subcellular environment gets remodeled during cellular perturbation. Applying our approach to characterize the cell biology of HCoV-OC43 infection, we demonstrated that profiling subcellular remodeling can reveal cellular responses not captured by abundance changes.

Expanding from previous methods developed for the native isolation of organelles,[20–23,25,26] our strategy uses CRISPR-Cas9-based endogenous tagging and circumvents the risk that overexpression might affect the enrichment of proteins in organelle IPs.[21] We show that native IPs provide rich composition profiles for both membrane-bound and membraneless organelles, demonstrating that this strategy can help characterize a wide range of cellular compartments. By using a broad collection of organelle markers, our dataset provides for each protein a complex signature of enrichment across many different IPs. This extensive coverage is particularly beneficial in the context of significant co-enrichment between organelles. A single IP might not provide enough information to fully disambiguate ER vs. mitochondrial proteins, for example. However, the distinctive enrichment profile of these two organelles across all pull-downs enables a clear differentiation. By analogy to gradient or differential fractionation studies, each native IP can be considered as a means to generate a targeted "fraction" of the cell. These "fractions" need not be pure. Rather, compartments get defined because their proteins share a distinctive enrichment pattern across the whole set of IPs and can be identified following the principle of correlation profiling.[9,10,38]

Our IP-based strategy extends the collection of methods developed for the proteome-wide characterization of subcellular localization.[12,14,16,18,27,28,31,36,53] An analysis focused on the mitochondrial proteome revealed a high degree of both precision and recall from our annotations. Furthermore, our dataset discriminates proteins from closely adjacent compartments, for example, Golgi vs. *trans*-Golgi, or early endosome vs. recycling endosome, which might be difficult to resolve using centrifugation-based methods because of similar biophysical properties and hence similar fractionation behaviors. Spatially adjoined compartments might also be hard to distinguish by proximity ligation; for example, outer mitochondrial membrane and peroxisomal proteins cluster together in a recent BioID-based map of the proteome.[18] Another advantage of our IP-based workflow is that sample preparation does not require complex instrumentation (e.g., ultracentrifuge), making it relatively simple to implement and scale. Given recent technological development in MS (for example, the increased sensitivity provided by ion mobility instruments,[104] or data-independent acquisition methods[37]), it is likely that the amount of input material needed for IPs could be significantly miniaturized. This would further improve scalability and application to cellular models for which large cell culture might be difficult, such as cells differentiated from stem cells. IP-based methods also enable the profiling of lipidomes[25] or metabolomes,[20,22,105] paving the way for a characterization of subcellular organization extending beyond proteins.

On top of providing a large data resource and a compendium of subcellular localization annotations, we also present a data-driven, graph-based analytical framework. This framework enables *de novo* identification of subcellular compartments and their interconnections and offers an unsupervised formalism to characterize subcellular remodeling. Our approach complements supervised methods that have been developed for the characterization of subcellular localization and dynamics using ground-truth markers.[106] In addition, we demonstrate that the k-NN representation of our dataset encapsulates, for each protein, specific information that can be mined to generate specific functional hypotheses. Our data-driven definition of interfacial proteins, including between compartments that are hard to resolve (for example, *trans*-Golgi and early endosomes), showcases how our approach can be leveraged for the analysis of interorganellar communication.

Finally, we provide a rich characterization of the pervasive subcellular remodeling induced by HCoV-OC43 infection, involving many different cellular compartments and pathways. By capturing responses across the entire cell, our work and others'[34,107] provide a window to understand the complex strategies that viruses have developed to concurrently hijack a diversity of cellular functions. Critically, our results suggest two orthogonal modes of regulation of host proteins, with distinct groups of proteins

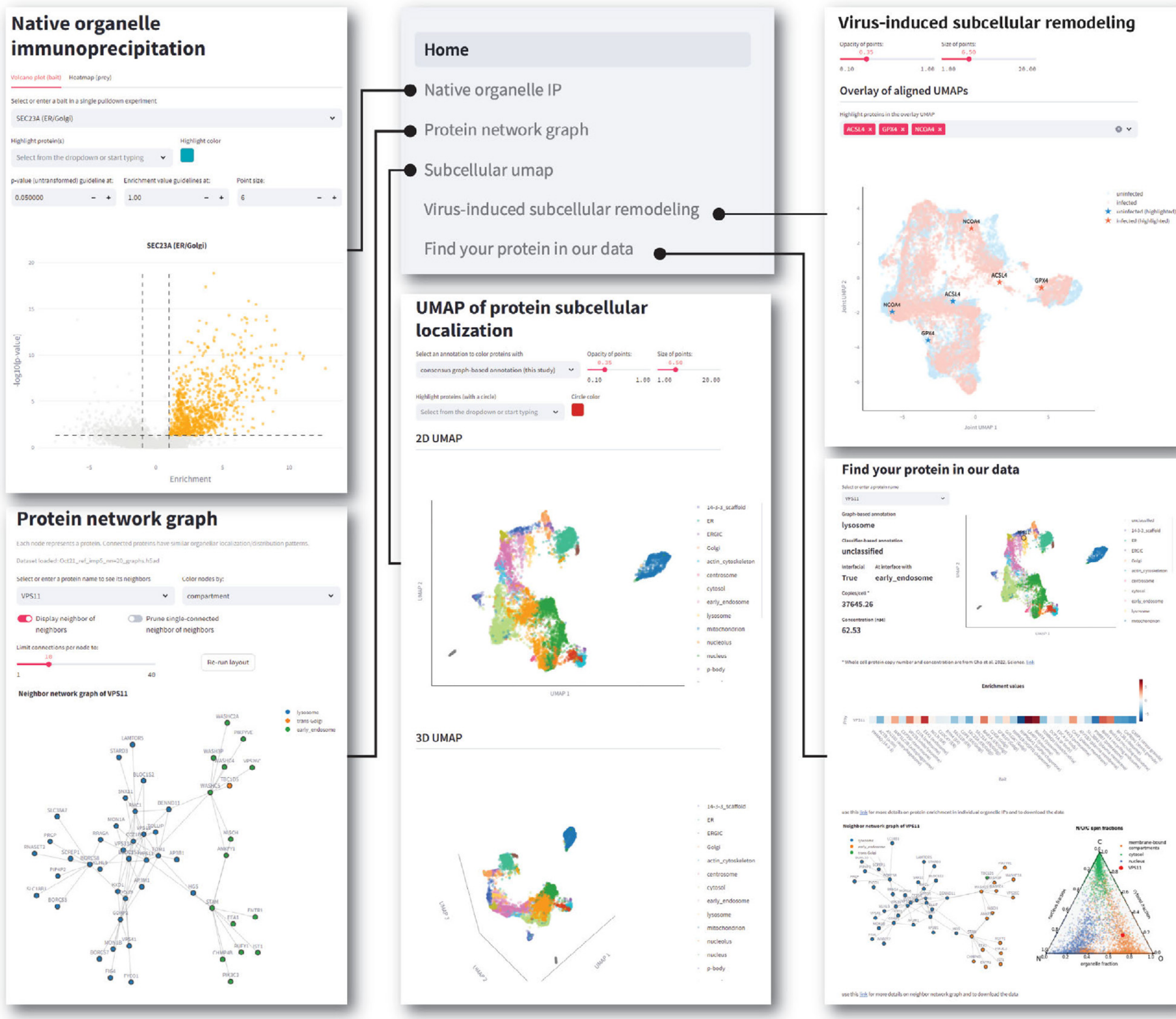

Figure 7. Interactive data exploration at organelles.czbiohub.org

regulated by changes in their subcellular environment or by changes in their overall abundance. A similar orthogonality between localization- and abundance-based regulation has been observed during proteome remodeling over the yeast cell cycle.[108] This underscores the importance of spatial regulation for the control of cellular functions, which is also exemplified by the ubiquity of protein re-localization in signal transduction.[7] Overall, in the same way that spatial methods are transforming transcriptome-based analyses,[109] scalable approaches that resolve the spatial organization of the proteome across different cell states have the potential to greatly extend our understanding of cellular functions in normal physiology and disease.

Limitations of the study

Because organelles are captured whole in native IPs, our method cannot resolve sub-organellar compartments. For example, our data do not distinguish membrane vs. lumenal proteins nor different mitochondrial or nuclear sub-compartments, which can be better discriminated by proximity ligation methods.[15,18] Our current workflow also requires the engineering of epitope-tagged cell lines. The development of organelle-specific antibody panels for capture could circumvent this limitation and expand the applicability of our method to untagged samples, for example, clinical specimen.[110] In addition, our approach does not address all aspects of subcellular localization. First, many proteins might localize to more than one compartment,[12,14] which is not captured by our current annotations. In the future, multi-label models could be trained on our data to predict multi-localization.[111] Second, subcellular proteomes are likely to change between different cell types. Systematically applying our methods to different cell types, for example, leveraging stem-cell-based differentiation systems,[112] would offer a solution. Third, we have not directly analyzed how splicing or PTMs influence the subcellular localization of different proteoforms.

PTM enrichment could be performed downstream of native IPs to provide more insights. Finally, our mass-spectrometry experiments are performed on bulk cell populations and cannot resolve variations of protein localization at the single-cell level. Subcellular localization can change as cells occupy different states (for example, across the cell cycle[108]), which are not discriminated in our population-average and steady-state measurements. Image-based methods might be better suited to capture single-cell localization heterogeneity and dynamic remodeling in live cells. Overall, these examples illustrate how the comprehensive characterization of subcellular architecture will likely require complementary methods applied in parallel.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Manuel Leonetti (manuel.leonetti@czbiohub.org).

Materials availability

Tagging plasmids have been deposited to Addgene (see key resources table). Engineered HEK293T cell lines endogenously tagged with organelle markers are available from the lead contact upon request.

Data and code availability

- Raw MS data have been deposited to the ProteomeXchange Consortium via the PRIDEpartner repository. RNA sequencing data have been deposited to the Gene Expression Omnibus (GEO). All data are publicly available; see accession numbers in the key resources table.
- All original code has been deposited to GitHub and Zenodo and is publicly available as of the date of publication. DOIs are listed in the key resources table.
- Any additional information required to reanalyze the data reported in this work is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: M.Y.H., D.P., K.K., C.L., D.N.I., J.E.E., and M.D.L.; methodology: M.Y.H., D.P., V.T., F.M., K.K., C.L., L.S., J.B., D.N.I., J.E.E., and M.D.L.; software: D.P., K.K., Y.A., and J.B.; investigation and validation: M.Y.H., D.P., V.T., F.M., K.K., C.L., L.S., C.J., R.B.-N., M.S., S.V., S.B., M.V., J.B., L.N., E.W., I.E.I., J.R.B., S.P., C.G.G., Y.A., J.S.C., A.H.M., S.S., B.C.D., D.N.I., J.E.E., and M.D.L.; writing – original draft: M.Y.H., D.P., C.L., J.E.E., and M.D.L.; writing – review & editing: M.Y.H., D.P., J.E.E., and M.D.L.; supervision and funding acquisition: S.Y.-L., S.B.M., D.N.I., J.E.E., and M.D.L.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-OC43 clone 541-8F (exact epitope unknown)	Millipore Sigma	Cat# MAB9012; RRID: AB_95424
Goat anti-mouse Alexa 488	Thermo Fisher Scientific	Cat# A-11001; RRID: AB_2534069
Bacterial and virus strains		
OC43 Betacoronavirus 1	ATCC	Cat# VR-1558
Chemicals, peptides, and recombinant proteins		
Recombinant S.p. Cas9-NLS purified protein	UC Berkeley Macrolab	N/A
Nedisertib (M3814)	Selleckchem	Cat# S8586
Pierce™ Anti-HA magnetic resin	Thermo Fisher Scientific	Cat# 88836
Lys-C Lysyl Endopeptidase	FUJIFILM Wako via Fisher Scientific	Cat# NC9223464
Trypsin protease, MS grade	Thermo Fisher Scientific	Cat# 90058
Fibronectin	Corning	Cat# 356008
FeRhoNox-1 (BioTracker 575 Red Fe^{2+} Dye)	Millipore Sigma	Cat# SCT030
Ferrostatin-1	Selleckchem	Cat# S7243
Liproxstatin-1	Selleckchem	Cat# S7699
RSL3	Yang and Stockwell[102]	Gift from James Olzmann, UC Berkeley
Critical commercial assays		
SF Cell Line 96-well Nucleofector Kit	Lonza	Cat# V4SC-2096
Deposited data		
Single-cell RNA Sequencing data of HCoV-OC43-infected HEK293T	This study	Gene Expression Omnibus (GEO): GSE278059
Mass spectrometry raw files and intensity tables	This study	ProteomeXchange: PXD046440
Experimental models: Cell lines		
HEK293T	ATCC	Cat# CRL-3216
Endogenously tagged HEK293T engineered cell lines, see Table S1	This study	N/A
Oligonucleotides		
Alt-R™ CRISPR-Cas9 crRNA, 2 nmol	Integrated DNA Technology	Sequences in Table S1
Alt-R™ CRISPR-Cas9 tracrRNA, 100 nmol	Integrated DNA Technology	Cat# 1072534
MULTI-Seq barcoded oligonucleotides	McGinnis et al.[113]	Gift from Zev Gartner, UCSF
Recombinant DNA		
All homology donor templates for endogenous tagging	This study	Sequences on Mendeley Data https://doi.org/10.17632/gmjf2wjkz4.1
Software and algorithms		
Detailed software for all data analyses and figure generation	This study	github.com/czbiohub-sf/Organelle_IP_analyses_and_figures; archived on Zenodo: https://doi.org/10.5281/zenodo.14000085
10x Genomics Cell Ranger v5.0.1	Zheng et al.[114]	https://www.10xgenomics.com/support/software/cell-ranger/downloads
MaxQuant v2.2.0.0	Cox and Mann[115]	https://www.maxquant.org

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
XGBoost v2.0.0	Chen and Guestrin[52]	https://github.com/dmlc/xgboost/releases/tag/v2.0.0
Enrichr API	Kuleshov et al.[116]	https://maayanlab.cloud/speedrichr
AlignedUMAP v0.5.2	McInnes et al.[47]	https://github.com/lmcinnes/umap
Cytoself	Kobayashi et al.[71]	https://github.com/royerlab/cytoself
scikit-optimize v0.9.0	https://zenodo.org/records/5565057	https://github.com/scikit-optimize/scikit-optimize/releases/tag/v0.9.0
matplotlib v3.5.3	The Matplotlib development team	https://matplotlib.org/
seaborn v0.12.2	https://joss.theoj.org/papers/10.21105/joss.03021	https://seaborn.pydata.org/whatsnew/v0.12.2.html
plotly v5.11.0	Plotly Technologies Inc	https://plotly.com/
networkx v3.3	NetworkX developers	https://networkx.org/documentation/stable/release/release_3.3.html
pysankey v1.4.1	https://github.com/Pierre-Sassoulas	https://github.com/Pierre-Sassoulas/pySankey/releases/tag/v1.4.1
python-ternary v1.0.8	https://zenodo.org/records/34938	https://github.com/marcharper/python-ternary/releases/tag/1.0.8
Upsetplot v0.9.0	Lex et al.[117]	https://github.com/jnothman/UpSetPlot/releases/tag/0.9.0
Other		
96wp nucleofector instrument	Lonza	Cat# AAF-1003S
Flow cytometry activated cell sorting	SONY	SH800S
23G blunt 1” needle	SAI Infusion Technology	Cat# B23-100
NormJ-ect-F 1mL syringe	Air-Tite	Cat# NJ-9166017-02
96w dipping magnet	V&P Scientific	Cat# VP 407AM-N1
Flat 96wp magnet	Thermo Fisher Scientific	Cat# AM10027
Confocal fluorescence microscope	Andor	Dragonfly
63x 1.47NA oil objective	Leica	N/A
EMCCD microscope camera	Andor	iXon Ultra 888
Cellview glass bottom 96wp, microscopy grade	Greiner Bio One	Cat# 655891
Orbitrap Mass spectrometer	Thermo Fisher Scientific	Fusion Lumos
C18 reverse phase chromatography columns	IonOpticks	Cat# AUR2-25075C18A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell culture

HEK293T cells (ATCC CRL-3216) were cultured in DMEM high-glucose medium (Gibco, cat. #11965118) with 10% fetal bovine serum (Omega Scientific, cat. #FB-11), supplemented with 2mM glutamine (Gibco, cat. #25030081), penicillin and streptomycin (Gibco, cat. #15140163). All cell lines were maintained at 37°C and 5% CO_2 and routinely tested for the absence of mycoplasma.

METHOD DETAILS

Cell culture & CRISPR/Cas9 engineering

Cell culture

HEK293T cells (ATCC CRL-3216) were cultured in DMEM high-glucose medium (Gibco, cat. #11965118) with 10% fetal bovine serum (Omega Scientific, cat. #FB-11), supplemented with 2mM glutamine (Gibco, cat. #25030081), penicillin and streptomycin (Gibco, cat. #15140163). All cell lines were maintained at 37°C and 5% CO_2 and routinely tested for the absence of mycoplasma.

Cell line engineering – tagged organelle markers

CRISPR/Cas9 methods were used for gene editing by homology directed repair (HDR) following established protocols. Ribonucleoprotein (RNP) complexes of *S. pyogenes* Cas9 and guide RNA were pre-assembled in vitro, mixed with double-stranded (dsDNA) HDR donors and delivered into HEK293T cells by electroporation in 96-well plates (see below). Each electroporation used 0.2×10^6 cells, 70 pmol RNP and 2 pmol HDR template. Electroporated cells were incubated for two days in the presence of 1 μM nedisertib (M3814; Selleckchem # S8586) to enhance HDR efficiency.[118] dsDNA HDR donors including tag sequence flanked by gene-specific homology arms were PCR-amplified from a corresponding plasmid template using 5' biotinylated PCR primers as previously described.[119] RNP complexes were freshly assembled prior to electroporation and combined with HDR template in a final volume of 10 μL. First, 0.7 μL gRNA (130 μM stock; source: Integrated DNA Technologies) was added to 2.2 μL high-salt RNP buffer {580 mM KCl, 40 mM Tris-HCl pH 7.5, 20% v/v glycerol, 2 mM TCEP-HCl pH 7.5, 2 mM MgCl2, RNAse-free} and incubated at 70°C for 5 min. 1.8 μL of purified Cas9 protein (40 μM stock in Cas9 buffer, ie. 70 pmol) was then added and RNP assembly was carried out at 37°C for 10 min. Finally, HDR template (2 pmol) and sterile RNAse-free H2O were added to 10 μL final volume. Electroporation was carried out in Amaxa 96-well shuttle Nucleofector device (Lonza) using SF solution (Lonza) following the manufacturer's instructions. Cells were washed with PBS and resuspended to 10,000 cells/μL in SF solution (+ supplement) immediately prior to electroporation. For each sample, 20 μL of cells (ie. 200,000 cells) were added to the 10 μL RNP/template mixture. Cells were immediately electroporated using the CM130 program, after which 100 μL of pre-warmed media (containing 1 μM nedisertib) was added to each well of the electroporation plate to facilitate the transfer of 25,000 cells to a new 96-well culture plate containing 150 μL of pre-warmed media (containing 1 μM nedisertib). Electroporated cells were cultured for >5 days and transferred to 12-well plates prior to selection by fluorescence-activated cell sorting (FACS). For each target, 1,200 cells from the top 1% fluorescent cell pool were isolated on a SH800 instrument (Sony biotechnology) and collected in 96-well plates.

Cell line engineering – localization de-orphaning

HEK293T cell lines used in the analysis presented in Figure 4 were engineered using the mNeonGreen_2(1-10/11) split fluorescent protein system using protocols previously described.[14] In brief, CRISPR/Cas9 methods were used for gene editing by homology directed repair (HDR) using RNP electroporation methods as described in the previous section, with the exception that single-stranded deoxyoligonucleotide (ssODN) donors were used (Ultramer ssODN, Integrated DNA Technologies; 120 pmol donor per electroporation).

Sample generation for proteomics analysis

Input material

All experiments were performed in triplicate using $\sim 10^7$ cells per replicate, harvested from 10cm-plate cultures grown to 80-90% confluency.

Cell homogenization

Prior to organellar immunoprecipitation or subcellular fractionation, HEK293T cells were mechanically homogenized by repeated passage through a blunt 23G needle in hypotonic buffer. Cells grown on 10cm plates were washed once in ice-cold PBS, and collected by scraping in 5 mL ice-cold PBS. All subsequent steps were conducted at 4°C or on ice. First, scraped cells were gently pelleted at 500 ×g (4°C) and resuspended in 950 μl of hypotonic homogenization buffer containing: 25 mM Tris-HCl pH 7.5, 50 mM sucrose, 0.2 mM EGTA, 0.5 mM MgCl2, complemented with protease and phosphatase inhibitor cocktail (Thermo Fisher Scientific, #PI78443). Resuspended cells were mechanically homogenized right away by passing the cell suspension 4 times through a 23G blunt 1'' needle (SAI Infusion Technology #B23-100) attached to a 1 mL syringe (Air-Tite NormJ-ect-F #NJ-9166017-02), followed by immediate thorough mixing with 89 μL of concentrated sucrose buffer (2.5 M sucrose, 0.2 mM EGTA, 0.5 mM MgCl2) to restore tonicity. Cell homogenates were clarified by centrifugation at 1,000 ×g for 10 min (4°C), resulting in a pellet containing mainly nuclear material, and a supernatant containing mainly organellar and cytosolic proteins.

Subcellular fractionation and whole cell extract preparation

For crude subcellular fractionation ("N/O/C" fractions, see text and Figure 1E), the clarified homogenate supernatant was further centrifuged at 20,000 ×g for 45 min (4°C), resulting in a pellet containing mainly organellar proteins, and a soluble cytosolic supernatant. Both the 1,000 ×g ("nucleus") and 20,000 ×g ("membrane-bound organelles") pellets were resuspended in SDS lysis buffer {5% SDS, 50 mM triethylammonium bicarbonate (TEAB) pH 7.55}, 750 μl and 500 μl, respectively. The cytosolic supernatant was supplemented with 20% SDS to yield 5% SDS final. All three fractions were boiled for 5 min. In parallel, a whole cell extract (WCE) was generated by directly scraping cells off a 10-cm plate in SDS lysis buffer. All N/O/C and WCE samples were further subjected to probe sonication (UP200St, Hielscher, Germany) and then clarified by centrifugation at 14,000 xg for 15 min. Protein lysates were quantified using BCA assay according to the manufacturer's instructions (Pierce, Thermo Fisher Scientific). 100 μg of protein was extracted by adding 4.5 volumes of ice-cold acetone, incubated at -20°C for 1 hour, and centrifuged at 21,000 xg at 4°C for 10 min. The protein pellet was then resuspended in {2.5% sodium deoxycholate, 50mM EPPS, pH 8.5} buffer. Protein samples were reduced by adding 1 mM DTT (Thermo Fisher Scientific, A39255) and incubating for 20 min at 37°C. Cysteine side chains were alkylated by adding 5 mM Iodoacetamide (IAA, Thermo Fisher Scientific #A39271) and incubated in the dark for 20 min at room temperature. Samples were digested overnight by adding Lys-C (Fujifilm Wako #NC9223464) at an enzyme-to-protein ratio of 1 mAU per 50 μg. Digestion continued the next day by adding trypsin (Thermo Fisher Scientific, 90057) at an enzyme-to-protein ratio of 1:100 for 3 h. All four steps were performed with shaking on a Thermomixer (Eppendorf, Germany). Digestion was stopped by acidification to 1% TFA, followed by incubation on ice for 5 min and centrifugation at 21,000 xg for 15 min to remove insoluble material.

Organellar IPs

The input material for organellar IPs is the clarified organellar/cytosolic supernatant (1,000 xg) from the cell homogenization step described above, prepared for cells expressing tagged organellar marker proteins. All steps were carried out at 4°C or on ice. For organelle capture, 750 μL of this supernatant were mixed with 20 μL Anti-HA magnetic beads (Thermo Fisher Scientific #PI88836) in a 96-deepwell plate (1 mL wells), and incubated for 10 min while shaking at 1,000 rpm in a thermomixer. Following capture, beads were bound to a dipping magnet (V&P Scientific #VP 407AM-N1) and washed 3 times in 150 μL ice-cold divalent-free D-PBS (Thermo Fisher Scientific #14190144) by serial transfers to 96-well plates. Beads were released in 150 μL ice-cold D-PBS in a 96-well plate, bound to a flat-bottom magnet (Thermo Fisher Scientific #AM10027), and the D-PBS supernatant removed. Beads were resuspended in 30 μL of DTT-urea buffer (2 M Urea, 12.5 mM Tris-HCl pH 7.5, 1 mM DTT). Bound proteins were alkylated by addition of 3.3 μL of 50 mM Iodoacetamide (IAA, Thermo Fisher Scientific #A39271) to each sample, followed by shaking (1,400 rpm) at RT for 30 min on a thermomixer. Proteins were digested by addition of 0.5 μg of LysC (Fujifilm Wako #NC9223464) and kept shaking overnight at RT. Finally, 1 μg of trypsin (Thermo Fisher Scientific #PI90058) was added, followed by shaking at RT for 4 h. Beads were removed and the peptides were acidified to a final concentration of 1% trifluoro-acetic acid (TFA).

Peptide desalting and mass spectrometry

Organellar IP, N/O/C, and WCE peptides were desalted on in-house prepared Styrenedivinylbenzene Reversed Phase Sulfonate packed Stagetips.[120] Briefly, Stagetips were activated with 100% methanol, conditioned with 80% acetonitrile/0.1% TFA, and equilibrated with 0.2% TFA, followed by sample loading, washing with 99.9% isopropanol/0.1% TFA, and additional washes: twice with 0.2% TFA and once with 0.1% formic acid. Organellar IP peptides were eluted with 60% acetonitrile/0.5% ammonium hydroxide. However, to achieve proteomic depth, N/O/C and WCE peptides were triple fractionated by first eluting with 40% acetonitrile 0.5% TFA 100mM ammonium formate, followed by 60% acetonitrile 0.5% TFA 150 mM ammonium formate and finally with 80% acetonitrile 1% ammonium hydroxide. Desalted peptides were flash frozen then dried by centrifugal evaporation and resuspended in 2% acetonitrile/0.1% TFA. Peptides were analyzed on a Fusion Lumos mass spectrometer (Thermo Fisher Scientific, San Jose, CA) equipped with a Thermo EASY-nLC 1200 LC system (Thermo Fisher Scientific, San Jose, CA) and nanoFlex ESI source. Peptides were separated by capillary reverse phase chromatography on a 25 cm column (75 μm inner diameter, packed with 1.6 μm C18 resin, AUR2-25075C18A, Ionopticks, Victoria Australia). Electrospray Ionization voltage was set to 2000 volts. Peptides were introduced into the Fusion Lumos mass spectrometer using a two-step linear gradient. For organellar IP samples, 3–27% buffer B (0.1% (v/v) formic acid in 80% (v/v) acetonitrile) for 52.5 min followed by 27-40% buffer B for 14.5 min. For N/O/C and WCE samples, 3–27% buffer B for 105 min followed by 27-40% buffer B for 15 min. Both operated with a flow rate of 300 nL/min. Column temperature was maintained at 50°C throughout the procedure. Data was acquired in top speed data-dependent mode with a duty cycle time of 1 s. Full MS scans were acquired in the Orbitrap mass analyzer (FTMS) with a resolution of 120 000 (FWHM) for organellar IP and 240 000 (FWHM) for N/O/C and WCE samples. Both with an m/z scan range of 375-1500 m/z. Selected precursor ions were subjected to fragmentation using higher-energy collisional dissociation (HCD) with quadrupole isolation window of 0.7 m/z, and normalized collision energy of 31%. HCD fragments were analyzed in the Ion Trap mass analyzer (ITMS) set to Turbo scan rate. Fragmented ions were dynamically excluded from further selection for a period of 60 seconds for organellar IP and 45 seconds for N/O/C and WCE samples. The AGC target was set to 1,000,000 and 10,000 for full FTMS and ITMS scans, respectively. The maximum injection time was set to Auto for both full FTMS scans and ITMS scans.

HCoV-OC43 infection

Virus stocks

OC43 was obtained from ATCC (VR-1558) and propagated in Huh7.5.1 cells grown in DMEM media at 34°C. Viral titers were determined by standard plaque assay using BHK-T7 cells. Briefly, BHK-T7 cells were seeded at 800,000 cells per well in 6-well plates and grown in DMEM media for 24 h at 34°C. The next day, serial 10-fold dilutions of virus stocks were used to infect cells for 2 hours at 34°C, after which media was removed, and a solution of DMEM media containing 1.2% Avicel RC-591 was overlaid onto the cultures. Infected cells were incubated for 6 days at 34°C, followed by fixation with 4% formaldehyde, staining with crystal violet, and plaque counting. All experiments were performed in a biosafety level 2 laboratory.

Infection – organelle IPs

For organelle IPs from infected cells, HEK293T cells expressing tagged organellar markers were seeded ~18 h before infection at a density of $\sim 5*10^6$ per 10 cm plate, resulting in a confluency of ~80% at the time of infection. Infection was carried out by direct addition of viral inoculum into the media, at a multiplicity of infection (MOI) of 0.25, at 37°C. This MOI was optimized to result in a uniformly infected population at 48 hpi while minimizing the required amount of inoculum. Cells were then harvested and processed like uninfected cells described above.

Microscopy imaging

Sample preparation

Live-cell imaging was performed on 96-well glass-bottom plates (Greiner Bio One, cat. #655891) coated with 50μg/ml fibronectin (Corning, cat. #356008). Cells were seeded on an imaging plate ~30 hours before imaging at 15,000 cells per well. Before imaging, cells were counter-stained with the live-cell DNA dye Hoechst 33342 (Invitrogen, cat. #H3570) by incubation for 30 minutes at 37°C in 150 μl of Hoechst diluted to 1μg/mL in culture media. Media was then replaced with phenol-free DMEM (Gibco, cat. #21063029) supplemented with 10% FBS. Hoechst staining was performed three to four hours prior to imaging to provide the cells time to recover from any mechanical stress due to medium changes. In experiments involving FeRhoNox-1 (Sigma-Aldrich, cat. #SCT030), FeRhoNox-1 was resuspended following manufacturer's protocol and used at a final concentration of 5 μM in combination with Hoechst. Cells were then incubated at 37ºC for 1.5 hours in serum-free imaging media, and subsequently washed and imaged in serum-free imaging media.

Live-cell fluorescence microscopy

Cells were imaged on a DMI-8 inverted microscope (Leica) equipped with a Dragonfly spinning-disk confocal system (Andor), a 63x 1.47NA oil objective (Leica), and a 16-bit iXon Ultra 888 EMCCD camera (Andor, pixel size: 13x13 μm2). A pinhole size of 40 μm was used with an EM gain of 400. Cells were maintained at 37°C and 5% CO2 during image acquisition by a stage-top incubator (Okolab, H101-K-Frame). The microscope was controlled using the open-source microscope-control software MicroManager (version 1.4.22).

RAB14 co-localization. A cDNA construct encoding RAB14 fused with mCherry at the N-terminus was cloned into an expression plasmid under CMV promoter expression. The construct was transfected into cells using FuGENE-HD following manufacturer's protocol (Promega cat. #HD-1000; 3:1 transfection reagent to DNA ratio; 72 hours incubation). Co-localization was quantified using the ROI manager in ImageJ.

Infection – imaging

The day prior to infection, cells were seeded on 96-well glass-bottom plates (Greiner Bio One, cat. #655891) coated with 50 μg/mL fibronectin (Corning, cat. #356008; manufacturer's protocol). 4000-8000 cells per well were seeded for uninfected conditions, and at least 8000 cells per well were seeded for infected conditions. On the day of infection, virus aliquots were thawed on ice, diluted in growth media, and applied onto cells at indicated MOI and at a final volume of 100 μL/well. For MOI calculations, cells were assumed to have tripled. MOI 0 conditions simply received media change. Unless indicated otherwise, infection imaging experiments were live-imaged at 48 hpi. All imaging experiments followed the sample preparation protocol described above.

ATG5 co-localization. HEK293T cells were endogenously tagged for both NCOA4 (N-terminal mNeonGreen; gRNA sequence TTTCTTTTAAAGGAGCAGTG) and ATG5 (N-terminal mScarlet; gRNA sequence GTATGTACTGCTTTAACTCC). Single-cell clones were isolated by flow cytometry based on double-positive mNeonGreen + mScarlet fluorescence. Four individual clones were grown and imaged upon OC43 infection as described above. Co-localization was quantified by determining the percent of NCOA4 positive pixels that were also ATG5 positive.

COP-I subunits localization. Cells were imaged upon OC43 infection as described above. Maximum-intensity projections of COP-I subunits were segmented using Otsu's method. The average distance from each fluorescent object's centroid to the nucleus boundary was then calculated for each cell.

Flow cytometry

70000 cells per well were seeded into 12 well plastic plates day prior to infection. At 48 h post MOI 1 infection (final volume of 1 mL/well), cells were resuspended in trypsin, quenched, and fixed in 4% paraformaldehyde at room temperature for 15 min. In experiments involving FeRhoNox-1 (see "Sample Preparation" section for protocol), cells were treated with FeRhoNox-1 prior to fixation. Cells were then washed, permeabilized, and blocked using Perm/Wash buffer (BD Biosciences, cat. #554723; >30 min at room temperature). For antibody staining, cells were treated with 1:200 OC43 antibody (Millipore Sigma, cat. #MAB9012; >1 h at room temperature), washed, treated with 1:500 Alexa Fluor secondary antibody (Thermo Fisher Scientific; >30 min at room temperature), and washed (all steps here in Perm/Wash buffer). Finally, cells were resuspended in FACS buffer (HBSS with no ions (Gibco, cat. #14175095) with 1% FBS (Omega Scientific, cat. #FB-11) and 25 mM HEPES (Gibco, cat. #15630080)) and analyzed using CytoFLEX (Beckman Coulter).

Treatment with ferroptosis inhibitors & activator: in infection experiments co-treated with Ferrostatin (Selleck Chemicals, cat. #S7243) or Liproxstatin (Selleck Chemicals, cat. #S7699), cells in 12 well plates were first treated with 500 μL/well of virus at the desired MOI, and then immediately treated with 500 μL/well of 2x desired drug concentration to reach a final volume of 1 mL/well. In RSL3 experiments (RSL3 was a gift from James Olzmann, UC Berkeley), RSL3 was added 24 hpi. The cells were fixed and assayed for OC43 at 48 hpi. To measure cell viability post drug treatment, cells were washed with PBS, trypsinized, and resuspended in 1 mL of growth media. 100 μL of each well was then counted using Cytoflex.

Transcriptomics

For transcriptomics of OC43-infected cells, we cultured and infected cells essentially as for all proteomics experiments, on 10 cm plates at 37°C. Cells were infected at ~80% confluency with an MOI of 1.0. We staggered a series of infections so that samples were harvested in parallel in an uninfected state and at 16, 24, and 48 hpi, respectively. Cells were harvested by trypsinization, quenched, and washed in PBS. We then used MULTI-seq[113] to encode the experimental conditions (i.e. uninfected or 16, 24, 48 hpi) before pooling all cells and loading them onto one lane of a 10x chip (NextGEM Single-Cell V(D)J Reagents Kit v1.1, 10x Genomics), aiming for recovery of 20,000 cells. We used the 5′ workflow to distinguish the different viral subgenomic RNAs.[121] We followed the manufacturer's recommendation for library preparation and sequencing, spiking in the MULTI-seq amplicons at ~8% of the gene expression library, at ~50,000 reads per cell. We processed the raw data with cellranger v. 5.0.1 using a custom transcriptome reference that combines the GRCh38 human genome and a manually curated reference for OC43. In addition to the default cellranger filtering, we removed cells with <2,000 UMIs, cells with >5% content of mitochondrial transcripts, as well as cells whose MULTI-seq barcodes indicated multiplets or could not be unambiguously assigned to one experimental condition, resulting in 12,910 cells in the final dataset. We used Leiden clustering to define three main cellular states, corresponding to uninfected cells, and cells with low and high viral loads, respectively. To assess differential gene expression between uninfected and infected cells presenting high viral load, we generated pseudo-bulk transcriptomes from sets of 100 randomly subsampled cells from either cell state.

Data analysis – mass spectrometry proteomics

Data availability

Mass spectrometry raw data and associated MaxQuant output tables were deposited to the ProteomeXchange Consortium via the PRIDE partner repository (accession PXD046440).

Raw Data Processing

Raw data were processed in MaxQuant[115] v. 2.2.0.0. We defined separate parameter groups for organellar IPs and used the match between runs feature and MaxLFQ[122] for label-free quantification (using 1 min. LFQ ratio count), each within those groups. Spectra were searched against human protein sequences (downloaded from Uniprot on 2021-07-17, canonical and isoform sequences) and 23 manually compiled HCoV-OC43 protein sequences (assembled based on the genome of HCoV-OC43 strain 1588, provided by ATCC). Briefly, a 1% false discovery rate was set at the peptide spectrum and protein matching with a minimum peptide length of 7, 2 missed cleavages were allowed, and enzyme was set to trypsin. Oxidation of methionine and protein N-terminal acetylation were set as variable modifications along with carbamidomethylation of cysteines set as a fixed modification.

Protein identification in native immunoprecipitation

Protein identifications were filtered, removing common contaminants, hits to the reverse decoy database as well as proteins only identified through post-translational modifications at specific sites. We used $\log_2$ MaxQuant LFQ intensities for all analyses. We required that each protein be quantified in all replicates from at least one native immunoprecipitation-mass-spectrometry (IP-MS) sample and in two replicates of whole proteome MS samples. Missing values were addressed by imputation using a Gaussian distribution, centered on the left tail of the global distribution of LFQ intensities. Specifically, the offset of the mean of this Gaussian distribution is 1.8 times the global standard deviation, while its standard deviation is fixed at 0.3 times the global standard deviation.

Protein enrichment analysis using affinity-enrichment mass spectrometry

Our strategy for enrichment analysis uses the established concept of affinity-enrichment mass spectrometry (AE-MS).[123] In AE-MS, rather than using a single negative control to measure protein enrichment profiles for a given IP-MS sample, enrichment is calculated relative to a larger set of unrelated IP-MS samples (that is, a larger set of samples that do not specifically capture the same set of proteins as the sample of interest). As previously described,[14,123,124] this enables a better estimation of the null distribution of background binders and leads to a more robust identification of interactions.

Sample grouping

For the current study, the full set of triplicate IP-MS samples was generated across 17 separate experimental batches, each batch containing on average 8 triplicate samples (or 24 individual IPs). For each batch, we first defined the background of proteins that bind or adsorb to the IP beads in a manner that is not 3xHA tag-specific; this non-specific background is defined by the LFQ intensities measured in a triplicate set of negative control IP-MS included in each batch, obtained using wild-type (untagged) cells. To minimize batch effects, we then grouped together batches that shared similar background intensity profiles using hierarchical clustering of these negative controls (see details on Github). This analysis defined 5 overall groups of samples.

Enrichment calculation

All enrichment calculations use triplicate sets of measurements, which will be subsequently described as "samples". Within each of the five sample groups, raw protein enrichment values were first computed using the cohort-matching, untagged negative control as the null distribution. The pairwise correlation of raw protein enrichment between all samples in a group was calculated, and uncorrelated samples were identified (correlation coefficient <0.35). Final protein enrichment values were then calculated using, for each sample, the full set of other uncorrelated samples from the same group to define a null distribution. The final enrichment value for each protein was calculated by taking the difference in median $\log_2$ LFQ values between the replicates and the null distribution. The significance of the final enrichment was tested using a t-test comparing the mean of $\log_2$ LFQ values between the replicates and the null distribution. The entropy of each protein's enrichment profile was computed using Shannon's entropy formula after sum-normalizing the profiles.

Calculation of protein proportions in N/O/C spin-fractions

Protein identifications in spin-fraction MS samples were filtered using identical methods outlined for IP-MS samples. We required that each protein be quantified in at least one out of three spin-fractions. The N/O/C proportions for each protein (from triplicate experiments) were defined by dividing median N, O or C LFQ values by the sum of median LFQ values obtained from all three spin-fractions.

Graph-based analyses

k-nearest neighbor (k-NN) graph

To construct the k-NN graph, we combined the protein enrichment data matrix with the N/O/C proportions data matrix. This integration required the presence of the same proteins in both matrices (that is, proteins absent from one of the matrices were excluded from analysis). We employed the *scanpy* package[125] for the construction of the k-NN graph. In this process, the combined data matrix was first scaled to achieve zero mean and unit variance with clipping values at 10. Subsequently, we applied the UMAP algorithm,[47] using 20 nearest neighbors and the Euclidean distance metric, to generate a weighted adjacency matrix delineating the edges of the k-NN graph.

Protein pairwise correlation/distance

Pearson correlation coefficient between each protein pair was computed from the scaled combined matrix. Cosine distance between each protein pair was computed using 30 principal components derived from the scaled combined matrix through principal component analysis.

Dimension reduction

The combined matrix of enrichment value and N/O/C proportions were scaled to achieve zero mean and unit variance with clipping values at 10. The UMAP algorithm[47] was used to embed the scaled matrix in two or three-dimensional space using 20 nearest neighbors, the Euclidean distance metric, and a minimum embedding distance of 0.1.

Clustering and protein-level compartment annotation

The clustering of protein groups was computed with the Leiden algorithm implemented in the *scanpy* package.[125] The k-NN graph was used as input and the "partition_type" parameter was set to "leidenalg.RBConfigurationVertexPartition". Protein-level compartment annotations were obtained through ontology enrichment analysis. This analysis was conducted using the Enrichr[116] API at https://maayanlab.cloud/speedrichr. For each Leiden cluster, enrichment was calculated against a background list including all proteins detected in our dataset. Enrichment analysis was computed using the COMPARTMENTS[39] and GO-Cellular component[50] databases using a p-value cutoff of 0.01. The most significantly enriched gene ontology term for each Leiden cluster was then assigned as the annotation for the proteins in that cluster. Leiden clusters with the same annotation were merged.

Clustering score

We used the Caliński and Harabasz index[126] (CHI) to measure the extent to which proteins sharing a given ground-truth label form well-delineated clusters. The CHI is a normalized measure of the ratio of distances between separate clusters (how well different clusters separate from each other in a plot) to the dispersion within each cluster (how tightly packed each cluster is). Higher CHI values represent better clustering resolution. The ground-truth labels were class-balanced, and the average CHI was computed over 200 random class-balancing sets to account for variability introduced by the process. Proteins observed by all datasets and included in the ground-truth sets were used to compute the CHI. For each dataset, UMAP embeddings were generated using ten random seeds. The CHI was then calculated based on organelle- and protein-complex-level ground truths. Finally, we reported the mean and standard error of the mean (SEM) for the index across class-balancing sets and UMAP random seeds.

Graph-based annotation

Protein-level compartment annotations were refined using the k-NN graph. For each protein, the compartment annotation most frequently observed among all its neighbors in the graph was adopted as the protein's graph-based annotation. This approach excluded the "unclassified" category from being selected as the most frequent annotation.

Jaccard coefficient

To calculate a Jaccard coefficient, two partites have to be defined to which a given node might be connected to. Therefore, for a given protein, we identified which two compartment annotations (partites) were the most found in the list of that protein's direct neighbors in the k-NN graph. Compartments comprising large numbers of proteins overall might have an advantage in this calculation. Therefore, to rank the top-two partite compartments for each protein while accounting for the size of different compartments, the number of that protein's neighbors annotated as belonging to a given compartment was normalized by the total size of that compartment. A minimum threshold was also established: to rank as one of the top-two partites, a compartment must contain three or more of the direct k-NN neighbors of given protein. Jaccard coefficients were computed by dividing the sum of neighbors in the two selected compartments by the sum of the size of the two compartments subtracting the sum of neighbors.

Cluster connectivity

Connectivity between any two given graph-based annotation clusters in the k-NN graph was quantified by calculating the percentage of connections between two clusters over the total possible connections that could exist between them.

Classifier-based analysis

We used XGBoost,[52] a classifier based on gradient boosting[127] that combines multiple decision trees to improve prediction performance. The compartment classifier was built using the eXtreme Gradient Boosting (XGBoost) algorithm.[52] implemented via the XGBoost package.[128] The training dataset comprises the combined matrix of enrichment values and N/O/C proportions. In the pre-processing stage, this matrix was scaled to achieve zero mean and unit variance with clipping values at 10. The training dataset comprised 61 IP-MS/spin-fraction-MS samples as features and 2412 literature-curated organelle markers (Table S2) as observation instances. To further address class imbalance, we employed a hybrid resampling technique known as SMOTENN,[129] which combines both over- and under-sampling. Hyperparameter tuning of the XGBoost model was conducted using the Bayesian optimization algorithm,[130] implemented via the *scikit-optimize* package (https://zenodo.org/records/5565057). The optimization process tuned the following parameters: ''max_depth'', ''learning_rate'', ''gamma'', and ''grow_policy'', using ''f1_weighted'' as the primary scoring metric. Three-fold cross-validation was applied in all model training processes (including hyperparameter tuning) to minimize the effect of randomly splitting training data into train and test subsets. During training, the model performance was monitored using ''mlogloss'', ''merror'', and ''auc'' metrics computed from the test subset. An ''early_stopping_rounds'' of 50 was configured to prevent model overfitting. The final evaluation of the model was performed with 5183 proteins that are not in the training dataset. Classifier-based annotations were derived utilizing an XGBoost model trained using a set of optimized hyperparameters (see Github for details). The ''objective'' parameter was set to ''multi:softprob'' to allow the generation of probabilistic predictions for each compartment. For each group of proteins, the compartment associated with the highest prediction probability was identified and designated as the classifier-based annotation.

Subcellular remodeling analysis

Aligned-UMAP

Protein identifications from IP-MS and N/O/C spin-fractions MS samples were separately processed for uninfected control and infected conditions following identical methods outlined above. After processing, the two combined matrices—one for the uninfected control condition and the other for the infected condition—were filtered to retain only the proteins shared by both matrices. As a result, two combined matrices were generated: one representing the uninfected control condition and the other for the infected condition. These matrices were then utilized to construct respective k-NN graphs. Both graphs were simultaneously embedded into a shared low-dimensional space in an aligned manner using the AlignedUMAP method, as implemented in the umap package (see umap-learn.readthedocs.io/en/latest/ aligned_umap_basic_usage.html). The embedding was configured with the following parameters: 20 nearest neighbors, Euclidean distance metric, minimum embedding distance 0.1, 300 epochs, and an alignment regularization factor of 0.002. To map viral proteins onto the infected side of the aligned UMAP, a separate UMAP was generated using the infected enrichment matrix including the viral proteins. The coordinates of the viral proteins were then projected onto the infected side of the aligned UMAP through Procrustes analysis.

Subcellular remodeling score

The uninfected and infected k-NN graphs were embedded into a shared 10-dimensional space using the AlignedUMAP method described above. To minimize stochasticity, we repeated the embedding process for 200 times. For each protein group, the remodeling score is defined as the average Euclidean distance between the infected and uninfected coordinates of 200 repetitions.

Visualization of protein subcellular remodeling

The uninfected and infected k-NN graphs were embedded into a shared 2-dimensional space for comparative visual representation. Protein-level compartment annotations were generated using the methodology outlined above, with a notable modification: the resolution parameter in the Leiden algorithm was adjusted to a value of 1.

Image analysis – deep learning with *cytoself*

A modified version of *cytoself*, an image-based organelle localization prediction method,[71] was used to determine orphan proteins localization (code released at github.com/jmhb0/cytoself). Specifically, a representation learning model trained on images from the OpenCell dataset[14] was used to identify proteins from OpenCell that had similar subcellular localization as the orphan proteins. Briefly, using max-intensity projections, nuclei were segmented using Cellpose[131] with default settings, and image crops were taken around each nucleus with height and width of 200 pixels, discarding crops with nucleus diameter smaller than 50 pixels. Each crop was made to have a nuclear channel, a protein channel, and a third channel where each pixel is the normalized distance to the nearest nuclear boundary[71] (where distance is negative for pixels in the nucleus). The protein and nucleus channels in each crop were independently normalized to range [0,1]. A *cytoself* model was then trained with OpenCell and orphan image crops using the official repository forked on October 10, 2023 with default parameters, and taking the checkpoint after 12 epochs of no loss decrease. Since some organelle classes in OpenCell are underrepresented, data resampling was implemented so that each organelle class appears in at least 5% of samples. This improves the representation quality for underrepresented classes like mitochondria and lysosome. The 'vqvec2'' vector (the second stage of vector quantization capturing fine-grained features[132]) was then extracted for all OpenCell and orphan images. This approach was done for each crop as well as for 90° rotations and flips of each crop, which improves the feature robustness.[133] The 'consensus embedding' was then obtained for each protein by taking the element-wise mean of all crop embeddings, which was then reduced to 200 dimensions using principal component analyses. For each orphan protein, the embedding space distances to literature-curated organelle marker genes (Table S2) were then computed using the Pearson correlation metric and grouped by organelles. For each organelle marker gene group, the correlation distance of the closest non-outlier gene (outliers defined as exceeding the quartiles by 1.5 times the interquartile range) was then used to compare across organelles. For interpretability, the correlation distances were converted to correlation scores.

Figure generation and general code availability

Data analysis for this study was conducted using the Python programming language. Figures were generated in Python using matplotlib, seaborn, plotly, networkx, pysankey, python-ternary or upsetplot packages. All computer code and reference data used to conduct the analysis and generate the figures in this manuscript can be found on GitHub at github.com/czbiohub-sf/Organelle_IP_analyses_and_figures. Code has been archived on Zenodo with the DOI: https://doi.org/10.5281/zenodo.14000085.

QUANTIFICATION AND STATISTICAL ANALYSIS

The statistical approach for each experiment is described in the corresponding STAR Methods section and or in the figure legends.

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Supplemental figures

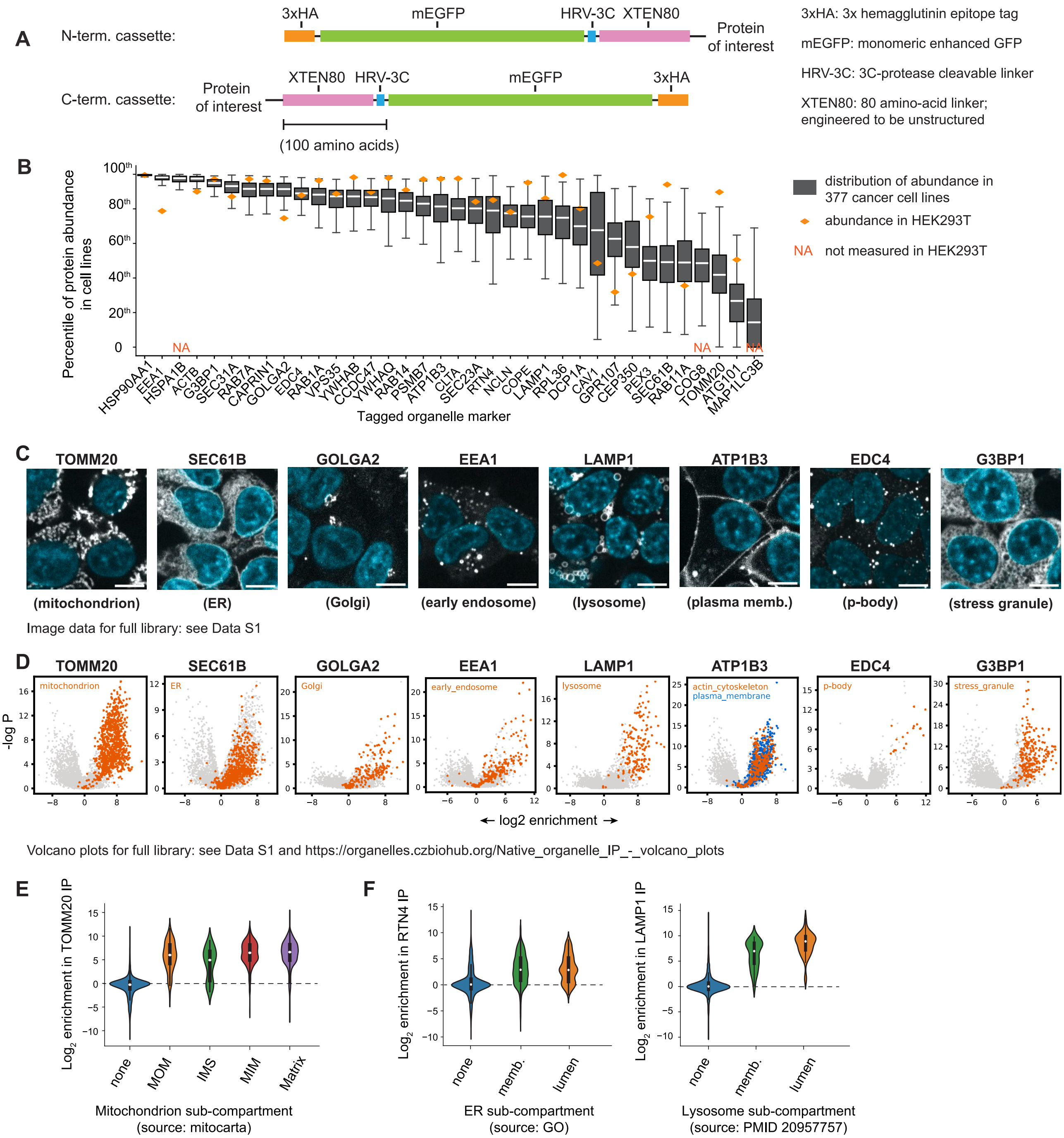

Figure S1. Tagged protein library, related to Figure 1

(A) N- and C-terminal tagging cassettes for organelle immunoprecipitations. See Table S1 for sequences.

(B) Protein abundance of all tagged targets in 377 cancer cell lines, expressed as percentile value from the proteome expressed in each cell line. 35 out of 37 targets (95%) are well expressed (median values above 40th percentile). Note that autophagy markers ATG101 and MAP1LC3B are only expressed in a subset of

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cell lines, which we attribute to autophagosomes not being universally present in cells but rather specifically produced when required. Boxplots show $25^{th}/50^{th}/75^{th}$ percentiles; whiskers show 1.5× interquartile range.

(C) Representative confocal fluorescence microscopy images of tagged organelle markers. See images for the full marker library in Data S1.

(D) Volcano plots showing protein enrichment in individual IPs. See plots for the full IP library in Data S1. Plots show $\log_2$ enrichment to unrelated controls (STAR Methods). *p* values: t test. Proteins annotated to localize to specific subcellular compartments are highlighted (graph-based annotations, see Table S4). IPs might also enrich for proteins from other compartments that are not highlighted. Note that the definition of each compartment in our annotations does not depend on any single IP, but rather is defined by proteins that share the same overall enrichment pattern across the whole high-dimensional dataset. As a result, it should not be expected that a single IP would enrich for all proteins annotated for a given compartment.

(E) Log_2 enrichment of specific categories of mitochondrial proteins in TOMM20 native IPs. MOM, mitochondrial outer membrane; IMS, intermembrane space; MIM, mitochondrial inner membrane; Matrix: mitochondrial matrix.

(F) Log_2 enrichment of membrane vs. lumenal proteins in RTN4 (endoplasmic reticulum) and LAMP1 (lysosome) IPs.

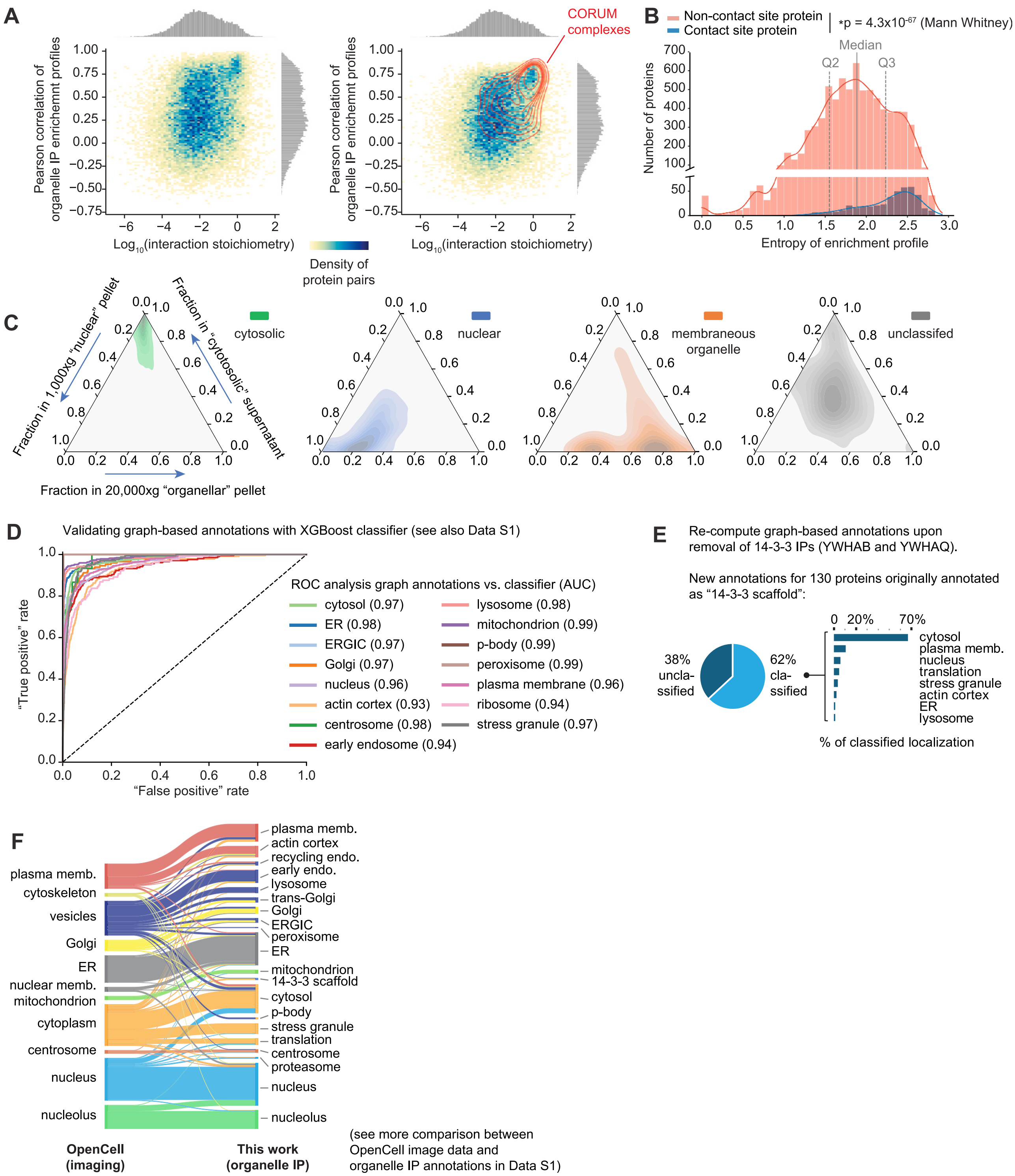

Figure S2. Protein enrichment signatures and annotation validation, related to Figure 2

(A) For each protein pair engaged in direct protein-protein interaction, plots show the relationship between interaction stoichiometry (x axis, data from the OpenCell database) and similarity of enrichment profile across the whole organelle IP dataset (y axis, showing Pearson correlation). Both panels display the same data; the right-hand panel highlights stable protein complexes annotated in CORUM.

(B) Distribution of enrichment entropy scores for contact site and non-contact site proteins. Contact site proteins were obtained from MCSdb.[46] Contact site proteins displayed higher entropy than non-contact site proteins (Mann-Whitney U test; $p = 4.27 \times 10^{-67}$).

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(C) Ternary density plots showing fractional abundance, for individual proteins, in N/O/C spin fractions (cf. Figure 1E). Different groups of proteins are shown in separate panels (from left to right): proteins annotated to localize in the cytosol (green), nucleus (blue), to membranous organelles (orange), and unclassified proteins (gray). While cytosolic, nuclear, or organellar proteins show very distinctive enrichment profiles in N/O/C fractions, unclassified proteins do not specifically enrich in any fractions (appearing in the center of the ternary plot), indicating their broad distribution across multiple compartments.

(D) Receiver operating characteristic (ROC) curves of graph-based subcellular localization annotations, using results from an XGBoost classifier as reference "ground-truth." Corresponding areas under the curve (AUC) values are shown.

(E) Graph-based annotations after removing YWHAB and YWHAQ IPs from the dataset. k-NN graphs were re-computed in the absence of enrichment data for YWHAB and YWHAQ IPs, and graph-based consensus annotations were obtained from direct neighbors (cf. Figure 2C, panel 5). Full list of proteins and annotations can be found in Table S3.

(F) Comparison of graph-based annotations with image-based annotations from OpenCell. In this comparison, only proteins annotated to localize to a single compartment in OpenCell are considered. See full list of proteins and annotations in Table S5 and additional comparison data in Data S1.

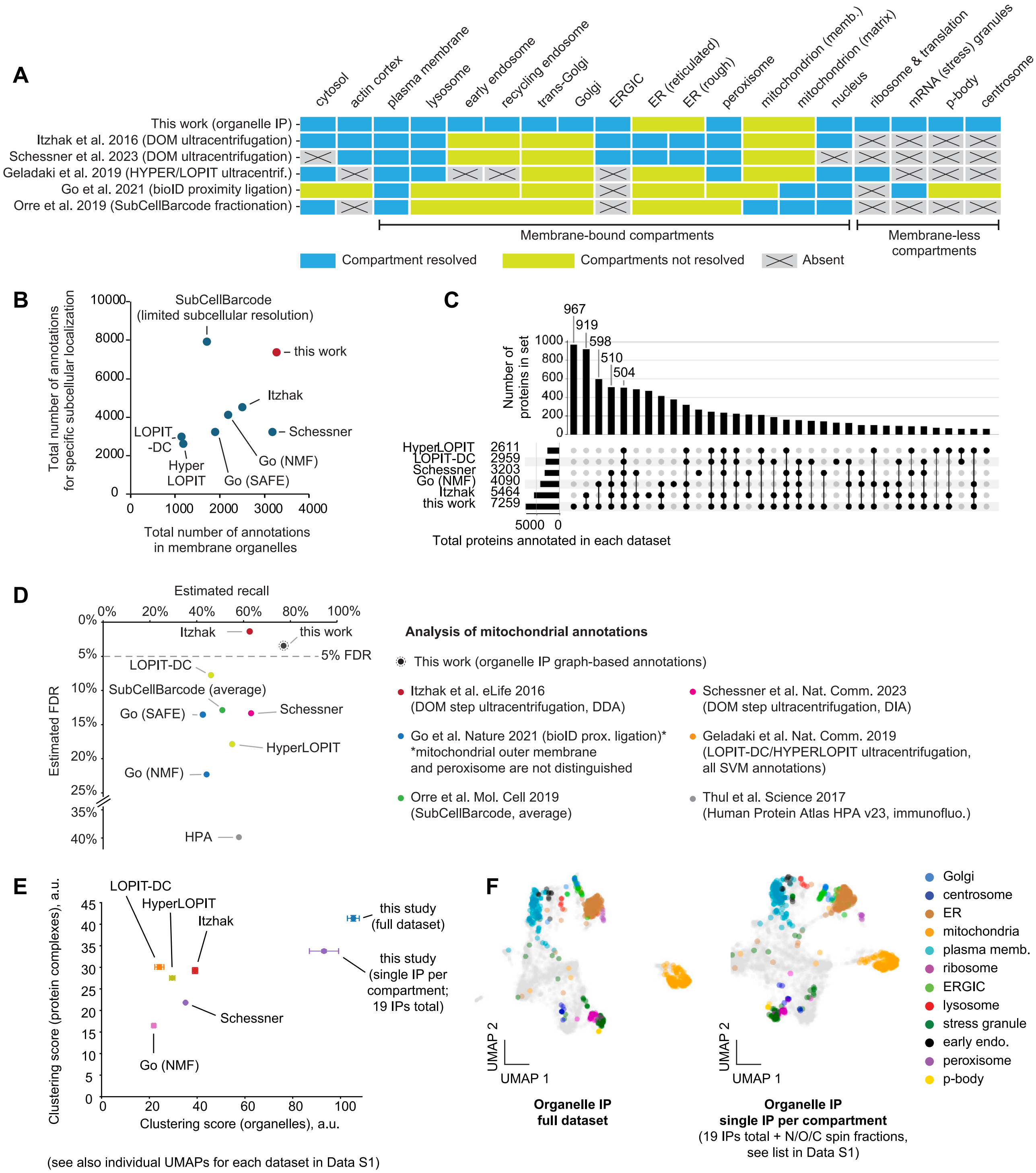

Figure S3. Comparison of annotations with existing datasets, related to Figure 2

(A) Comparison of the classes of subcellular compartments resolved in the annotations reported in large-scale mass spectrometry-based datasets using different proteomics methods: Itzhak et al.,[36] Schessner et al.,[37] Geladaki et al.,[53] Go et al.,[18] and Orre et al.[27]

(B) Comparison of the number of proteins annotated in different datasets. "Membrane organelles" refers to any membrane-bound compartment except the nucleus. In this analysis, only annotations to specific subcellular compartments were considered. This excludes any fully unclassified proteins, "large protein complex" annotations in Itzhak et al. and Schessner et al., and "secretory, unclassified" annotations in SubCellBarcode. Furthermore, protein isoforms are not considered: for proteins that share the same gene name, only a single annotation is considered. Proteins that do not have an explicit gene name (i.e., non-canonical protein products) were not included. "Curated" annotations were used for Geladaki et al.[53] (LOPIT-DC/HyperLOPIT).

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(C) Number of proteins annotated in different datasets. In this analysis, only annotations to specific subcellular compartments were considered, as in (B). Annotations from the SubCellBarcode dataset, which has limited subcellular resolution, were not included.

(D) Analysis of annotations for mitochondrial proteins. Estimation of recall and false discovery rate (FDR) of mitochondrial annotations in different large-scale datasets. A ground-truth reference of mitochondrial proteins is defined using the MitoCarta and MitoCOP databases (see text for details). For each dataset, all reported mitochondrial annotations were used (before any curation that might have used MitoCarta or MitoCOP), except for HPA, for which "uncertain" antibody annotations were not considered. See Table S5 for data. In this analysis, we defined as "unambiguously mitochondrial" the set of proteins annotated as mitochondrial in both MitoCarta AND MitoCOP databases (n = 990). The estimated recall is the fraction of these 990 proteins annotated as mitochondrial in the different large-scale datasets. We also defined as "very likely not mitochondrial" the set of proteins absent from BOTH MitoCarta and MitoCOP databases. The estimated FDR is the fraction of these proteins annotated as mitochondrial in the different large-scale datasets.

(E) Clustering scores. We measured two clustering scores capturing protein co-localization at different scales (see STAR Methods): an organelle-level score, using annotations curated from the literature (Table S2), and a protein-complex score, using annotations from CORUM.[55] The clustering scores measure the extent to which proteins sharing a given ground-truth label form well-delineated clusters (see individual UMAPs in Data S1). Only proteins that are annotated in all datasets were considered for clustering score calculation. We also report the clustering scores for a reduced version of our dataset in which only enrichment from 19 single IPs was used (one IP per compartment class from Figure 1D). Error bars represent SEM of clustering scores calculated over 200 class-balancing sets using 10 UMAP random seeds (STAR Methods). Some are smaller than the symbols.

(F) UMAPs comparing our full dataset to its reduced version (one IP per compartment class). Organellar marker proteins curated from the literature (Table S2) are labeled.

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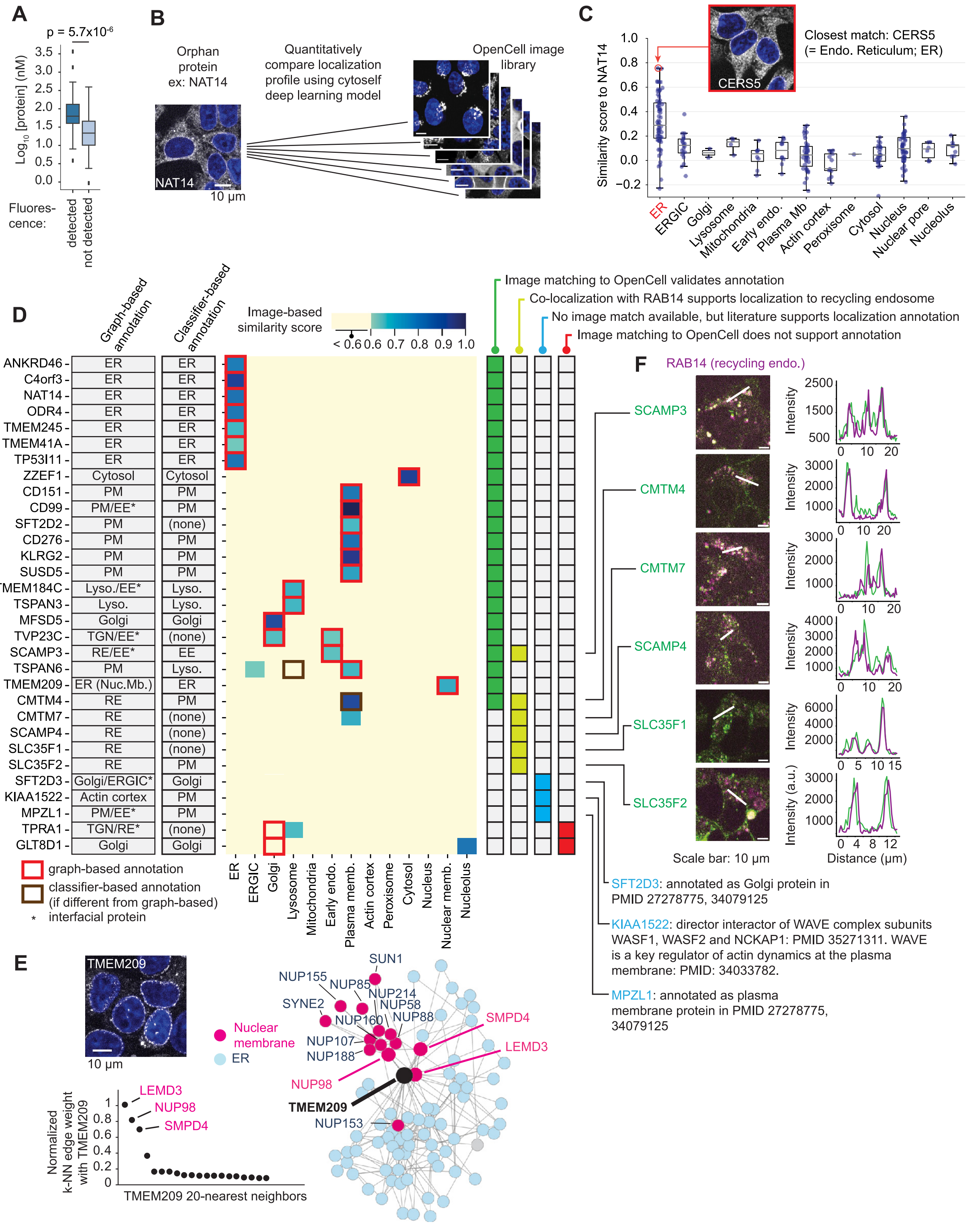

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Figure S4. De-orphaning subcellular localization—image analysis, related to Figure 4

(A) Protein abundance of tagged proteins. Tagged proteins from the 31 cell lines that displayed enough fluorescence to be selected by flow cytometry ("detected") displayed a significantly higher expression level than tagged proteins from the 51 cell lines for which fluorescence could not be detected (see Figure 4A). Boxplot shows $25^{th}/50^{th}/75^{th}$ percentiles; whiskers show 1.5× interquartile range. p value: Student's t test.

(B) Quantitative image analysis strategy: the subcellular distribution of each tagged orphan protein is compared with organelle markers found in the OpenCell image collection (using a literature-curated list of markers, see Table S2). The *cytoself* deep learning model was used to encode images into latent space embeddings. For each pairwise comparison of orphan vs. marker proteins, the Pearson correlation between embeddings provides a score that quantifies the similarity between the proteins' subcellular distributions.

(C) Distribution of similarity scores between the orphan protein NAT14 and the collection of OpenCell organelle markers. The highest similarity score is with the ER protein CERS5 (score: 0.76), validating our annotation of NAT14 as an ER protein (right).

(D) Heatmap representing, for each orphan protein, the maximum similarity score found with markers in specific organelle classes. Only similarity scores above 0.6 are reported. Abbreviations are as follows: ER, endoplasmic reticulum; PM, plasma membrane; EE, early endosome; lyso, lysosome; RE, recycling endosome; Nuc. Mb., nuclear membrane; ERGIC, ER-Golgi intermediate compartment.

(E) Connections of TMEM209 in the k-NN graph support its localization at the nuclear membrane. Shown are the normalized edge weights with TMEM209's 20 nearest neighbors in the k-NN graph.

(F) Co-localization of orphan proteins (mNeonGreen endogenous tag, green) with the recycling endosome marker RAB14 (mScarlet-RAB14 transient transfection, magenta) imaged by confocal microscopy. Fluorescence intensities along the white lines drawn on individual images are shown on the right panel.

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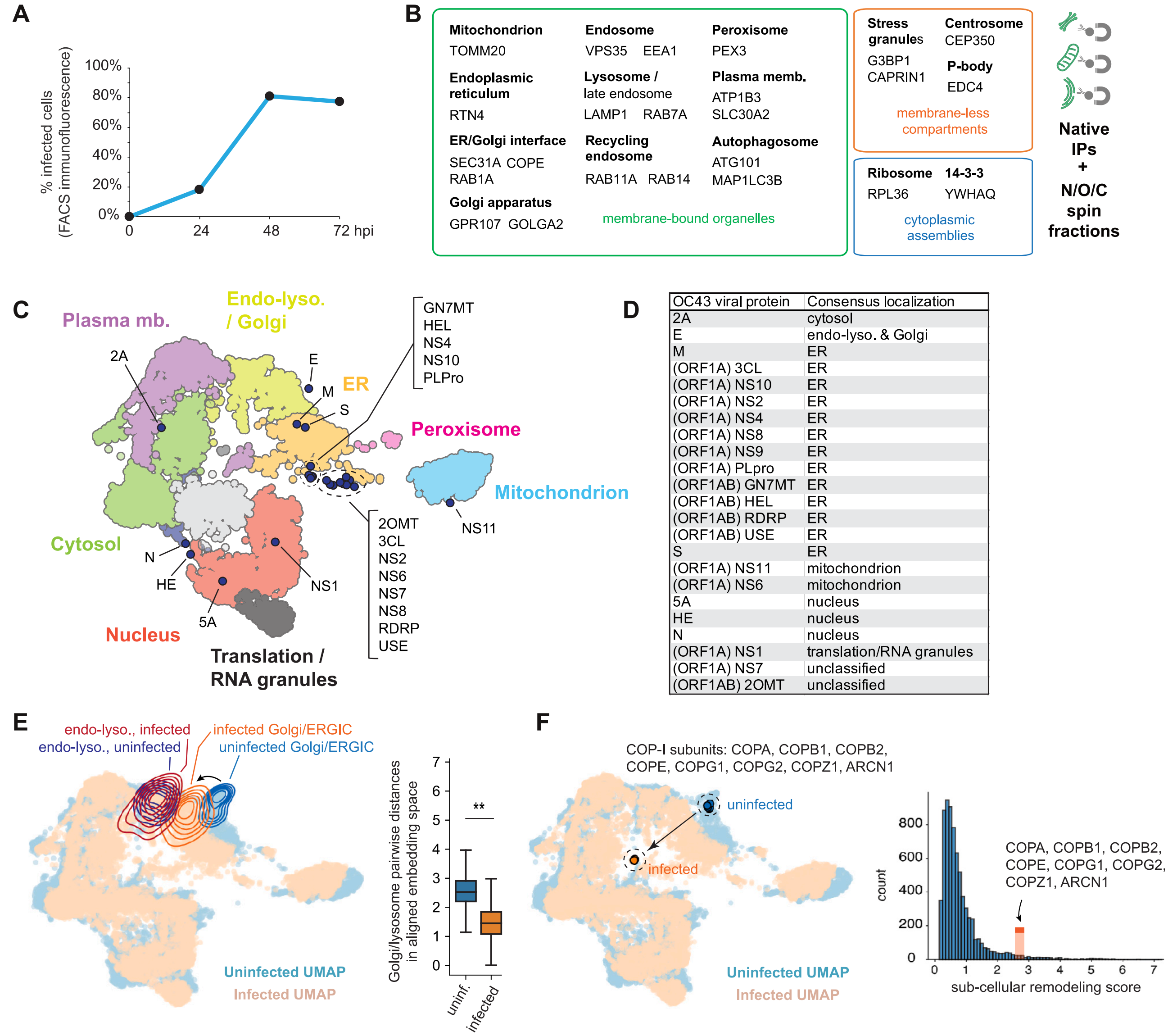

Figure S5. Subcellular remodeling during OC43 infection, related to Figure 5

(A) Percentage of OC43-infected cells as a function of time (hpi, hours post infection). Cells were infected at a MOI of 0.25 at the beginning of the experiment. Infected cells were detected by immunofluorescence flow cytometry using the monoclonal anti-OC43 antibody 541-8F, exact epitope unknown.

(B) Set of organelle IP markers included in the OC43 infection experiments. Triplicate IPs were processed side by side for uninfected and infected (48 hpi MOI = 0.25) samples for each marker. N/O/C spin fractions were also processed under uninfected and infected conditions.

(C) Two-dimensional UMAP of OC43-infected organelle profiling dataset showing subcellular territories defined by low-resolution Leiden clustering. The positions of virally encoded proteins are shown in the map.

(D) Annotated subcellular localization of virally encoded proteins, using the neighborhood-consensus annotation method shown in Figure 2C panel 5.

(E) Distances between Golgi and lysosomal proteins in aligned embedding space are significantly decreased upon infection ($p < 0.01$, Student's t test). This indicates that the organellar enrichment profiles for Golgi and lysosomal proteins become more similar upon infection. Boxplot shows 25th, 50th, and 75th percentiles; whiskers show 1.5× interquartile range.

(F) Subcellular remodeling profile of the COP-I subunits detected in our dataset.

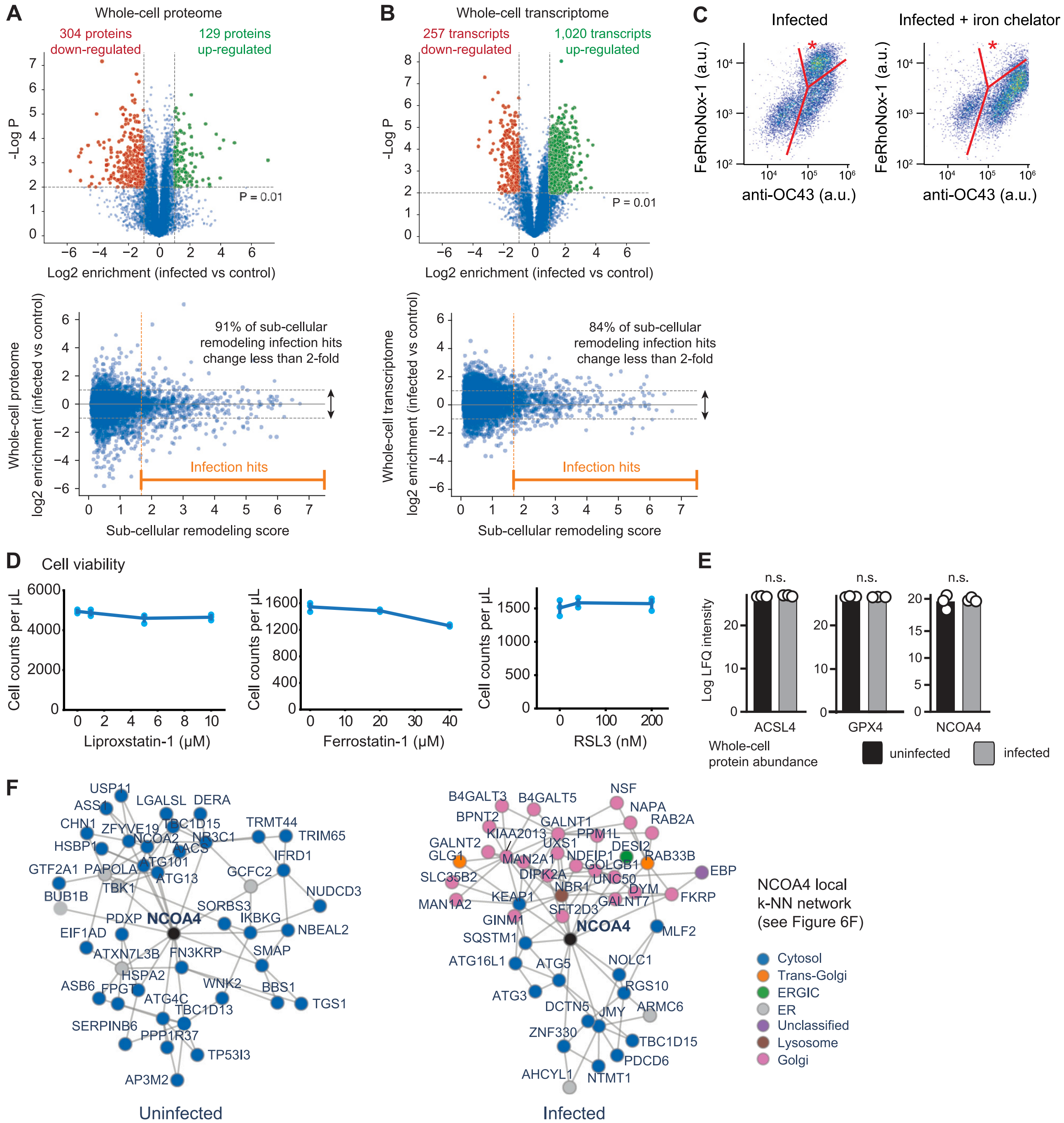

Figure S6. Abundance changes and ferroptosis activation during OC43 infection, related to Figure 6

(A) Whole-cell protein abundance changes upon OC43 infection. Top: volcano plot shows $\log_2$ enrichment of individual proteins in infected vs. uninfected whole-cell samples. *p* values are calculated from a t test using triplicate observations of each sample. Bottom: $\log_2$ enrichment of individual proteins in infected vs. uninfected whole-cell samples is plotted as a function of subcellular remodeling scores (cf. Figures 5A and 5B).

(B) Whole-cell transcript abundance changes upon OC43 infection. Top: volcano plot shows $\log_2$ enrichment of individual transcripts in infected vs. uninfected whole-cell samples. *p* values are calculated from a t test using triplicate observations of each sample. Bottom: $\log_2$ enrichment of individual transcripts in infected vs. uninfected whole-cell samples is plotted as a function of subcellular remodeling scores of corresponding proteins (cf. Figures 5A and 5B).

(legend continued on next page)

(C) Flow cytometry quantification of OC43 protein levels (anti-OC43 monoclonal antibody 541-8F, exact epitope unknown) and FeRhoNox-1 (5 μM, ferroptosis reporter). Treatment with the iron chelator nitroxoline (12 μM) led to a significant decrease in the number of cells positive for both OC43 and FeRhoNox-1 (subpopulation marked by an asterisk). Infection: MOI = 1, 48 hpi.
(D) Measurements of cell viability. Symbols show independent triplicate experiments, bars show SEM (some might be smaller than symbols).
(E) Whole-cell protein abundance measured by bulk proteomics.
(F) Local k-NN networks centered on NCOA4 in both infected and uninfected conditions. See also Figure 6F.